



EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Course Code: MENG 375
Course Name: Machine Elements
Semester/ Year: Fall 2024-2025
Instructor: Prof. Dr. Babak Safaei
Assistant: Mr. Emmanuel
Submission Date: 28/12/2024

Project Title: Design and Analysis of fracture behaviour of composite plate in 4 different thickness

Group : 2

STUDENT NO

22700513

23704019

21000040

21902496

22701924

NAME, SURNAME

Yanal Afaneh

Tarik Tariq

Yaşar Eren Cümelli

Hani fwzi

Haoran Wu

SECTIONS:

	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Abstract	1,6	5	5		
Introduction	1,6	5	10		
Analytical models	1,6	5	20		
Finite element analysis	1,6	5	25		
Discussion	1,6	5	15		
Conclusion	1,6	5	10		
Report Format And References			15		
TOTAL:(Out of 100)					

Instructions:	Student Outcomes:	
- No Late Submissions accepted. - Please include the equations, diagrams, plots, and figures in the report. - All the sources of data, equations, diagrams, plots, and figures used must be cited. - Please provide the references and cite each reference in the text. - All references must be cited in APA format. - Please submit the pdf soft copy of the report via email . - The file uploaded should be named as [MENG375-Project-Group_No] Files with any other name or format will be disregarded.	1	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
	2	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
	3	an ability to communicate effectively with a range of audiences
	4	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
	5	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
	6	an ability to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions
	7	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Abstract

This report investigates the fracture behaviour of composite plates with four different thicknesses to evaluate the effect of thickness on fracture resistance under various loading conditions. Composite materials, widely used in aerospace, automotive, and structural industries due to their superior strength-to-weight ratio, require a thorough understanding of their fracture mechanics across different thicknesses. Plates of 0.3 mm, 0.5 mm, 1 mm, and 2 mm thickness were analysed using numerical simulations to assess their response to applied stresses. The study focused on key parameters such as stress intensity factors (SIF), energy release rates, and crack propagation paths. Finite Element Methods (FEM), including the Extended Finite Element Method (XFEM) and cohesive zone models, were employed to simulate crack behaviour and growth, using the ANSYS platform for computational modelling. Results revealed that thinner plates experienced greater stress concentrations and lower crack propagation resistance, while thicker plates demonstrated improved energy dissipation and delayed fracture initiation due to enhanced fracture toughness. These findings underscore the trade-off between weight reduction and structural integrity in composite design. By providing critical insights into the relationship between plate thickness and fracture resistance, this study contributes to optimizing composite structures for improved safety and reliability in demanding engineering applications.

Keywords: Composite plates, Four different thicknesses, Fracture resistance, Under various loading conditions, Finite element methods, ANSYS.

Table of Contents

LIST OF FIGURES.....	4
LIST OF TABLES	5
INTRODUCTION	6
THEORY	14
MODEL.....	15
SIMULATION	15
SOLUTION	19
DISCUSSION	37
CONCLUSION	41
REFERENCES.....	42

List Of Figures

Figure 1 The different Crack behaviour of carbon fiber and modified composite carbon fiber	11
Figure 2 The three basic loading modes of crack are (a) Mode I (b) Mode II (c) Mode III	13
Figure 3 Material Properties	16
Figure 4 2D Sketch of the Composite Plate	16
Figure 5 Extruded section of the drawing.....	17
Figure 6 Image showing the conditions considered during the analysis.....	17
Figure 7 Figure showing the assignment of the coordinate system	17
Figure 8 Semi-elliptical crack details.....	18
Figure 9 Tetrahedron mesh on the plate.....	18
Figure 10 Tetrahedron mesh on the semi-elliptical crack.....	18
Figure 11 Outline tree	19
Figure 12 K1 value for 0.3 mm thickness.....	20
Figure 13 K2 value for 0.3 mm thickness.....	21
Figure 14 K3 value for 0.3 mm thickness.....	21
Figure 15 K1 value for 0.5 mm thickness.....	21
Figure 16 K2 value for 0.5 mm thickness.....	22
Figure 17 K3 value for 0.5 mm thickness.....	22
Figure 18 K1 value for 1 mm thickness	22
Figure 19 K2 value for 1 mm thickness	23
Figure 20 K3 value for 1 mm thickness.....	23
Figure 21 K1 value for 2 mm thickness	23
Figure 22 K2 value for 2 mm thickness	24
Figure 23 K3 value for 2 mm thickness	24
Figure 24 Stresses on X,Y,Z with time for 0.3 mm thickness	25
Figure 25 Schematic stress representation of 0.3 mm thickness on uniaxial load	25
Figure 26 Stresses on X, Y, Z with time for 0.5 mm thickness	26
Figure 27 Schematic stress representation of 0.5 mm thickness on uniaxial load	26
Figure 28 Stresses on X,Y,Z with time for 1 mm thickness	27
Figure 29 Schematic stress representation of 1 mm thickness on uniaxial load	27
Figure 30 Stresses on X,Y,Z with time for 2 mm thickness	28
Figure 31 Schematic stress representation of 2 mm thickness on uniaxial load	28
Figure 32 Strain graphs on X, Y, Z components for 0.3 mm thickness.....	29
Figure 33 Schematic strain representation of 0.3 mm thickness.....	30
Figure 34 Strain graphs on X, Y, Z components for 0.5 mm thickness.....	30
Figure 35 Schematic strain representation of 0.5 mm thickness.....	31
Figure 36 Strain graphs on X, Y, Z components for 1 mm thickness.....	32
Figure 37 Schematic strain representation of 1 mm thickness.....	32
Figure 38 Strain graphs on X, Y, Z components for 2 mm thickness.....	33
Figure 39 Schematic strain representation of 2 mm thickness.....	33
Figure 40 Crack extension graph with time for 0.3 mm thickness plate.....	34
Figure 41 Crack extension probe for every 0.2 seconds on 0.3 mm thickness plate	35
Figure 42 Crack extension graph with time for 0.5 mm thickness plate.....	35
Figure 43 Crack extension probe for every 0.2 seconds on 0.5 mm thickness plate	35
Figure 44 Crack extension graph with time for 1 mm thickness plate.....	35
Figure 45 Crack extension probe for every 0.2 seconds on 1 mm thickness plate	36
Figure 46 Crack extension graph with time for 2 mm thickness plate.....	36
Figure 47 Crack extension probe for every 0.2 seconds on 2 mm thickness plate	36
Figure 48: Thickness and Stress intensity factor graph	37
Figure 49: Thickness vs Max Elongation graph	38
Figure 50: Thickness and Maximum stress graph.....	38
Figure 51: Thickness vs Max strain graph.....	39

List Of Tables

Table 1 Each thickness with their collected data.	36
--	----

Introduction

Fracture mechanics is a branch of engineering that uses mathematical models to determine when a part with a crack or defect will break, particularly under extreme stress. This technique is critical for ensuring that engineering parts are strong enough to perform safely. **Effectively designing gradient structures** allows precise control over material properties, optimizing the balance between strength and toughness, and improving material safety during use [1]. Fracture toughness, which measures a material's ability to resist crack propagation, is a key factor in assessing material safety in service [1]. Understanding these properties is vital for evaluating materials' performance under real-world conditions.

Specimen-size-independent **fatigue crack growth** data is essential for analyzing fatigue failures from a theoretical and practical perspective. This data reflects a material's true resistance to fatigue crack growth, and using it is crucial for accurate structural modeling, especially when mechanical structures vary in size. Within the current fracture mechanics framework, **fatigue crack growth (FCG)** correlations, based on the ΔK approach, are often assumed to be unaffected by specimen size or shape [2]. However, the possibility of size dependence in crack growth is highlighted by geometric correction factors (GCFs), which are used in calculating the stress intensity factor (K). For instance, **IT** has shown through numerical analysis that the stress intensity factor depends on the specimens' H/W ratio (height-to-width ratio)[2]. An increase in thickness leads to a rise in the characteristic fracture strength. The characteristic fracture strength is at its lowest.[3]

We know that the Material cracks cause damage, reducing mechanical properties like the axial modulus under tensile loading and the shear modulus under shear loading, depending on crack orientation.

However, if the loading direction is reversed, these moduli can be restored [4]. After a **fatigue crack** initiates, it progresses through the small fatigue crack (SFC) phase, then the long crack (LC) phase, ultimately leading to transient fracture. Studies have found that crack initiation and small crack propagation account for 70-80% of the total fatigue life, making SFC growth behavior a critical aspect in predicting material service life and developing damage tolerance strategies [5]. Currently, fatigue performance evaluation is primarily based on the nominal stress–life (S-N) curve derived from fatigue tests. Building on this, methods such as the hot spot stress method, effective notch stress method, and structural stress method offer a more detailed understanding of local stress characteristics.[6].The proposed fatigue damage model, based on friction energy, is founded on a clear fatigue damage mechanism and requires only two independent parameters. The model was validated by comparing its predictions with the results of compression cycle tests conducted on concrete specimens.[7].The dynamic grid finite element method that tracks the free surface is too complicated and too fragile compared to fixed grid methods that capture the interface implicitly. This highlights the complexity of modeling crack propagation and material behavior under different stress conditions.[8].The Extended Finite Element Method (XFEM) is employed here to account for the displacement discontinuity. Various aspect ratios of elliptical voids are analyzed under both uniaxial and biaxial deformation boundary conditions. The findings indicate that voids with the largest surface area exhibit the highest growth rate compared to those with smaller surface areas."[9] Some fracture tips may not propagate even if they satisfy the Griffith criterion, as the global system seeks to minimize energy and form a stable crack pattern. The double virtual incremental method is an effective approach to assess whether the group of tips should advance.[10].

moreover, the use of **fiber- and textile-reinforced cementitious composites** has surged in recent decades due to their excellent mechanical properties, high shape flexibility, and relatively good durability. These composites are particularly valuable in both new structures and the reinforcement of existing ones. Thin-walled structures made from these composites provide effective protection against dynamic and extreme loading conditions [11]. Carbon fibers (CF) are widely used for multifilament rovings due to their excellent performance-to-cost ratio and resistance to aggressive environments. However, their **brittleness** and surface defects present challenges, especially under pull-out forces or when embedded in matrices containing sharp particles. To enhance the fibers' stability and longevity, polymer sizing and yarn impregnation techniques are employed [11]. Quantifying the fracture toughness of cement paste is challenging, both through standard nanoindentation tests and through the time-consuming and costly process of measuring micro-specimens.[12]

Polymeric-impregnating agents, such as **epoxy, polyacrylate, and styrene-butadiene**, are often used to improve the mechanical properties of textile-reinforced composites (TRC). However, they can weaken the bond between fibers and substrates and reduce fire resistance, particularly at high temperatures. To address these issues, **inorganic impregnating agents**, like amorphous silica, have been developed to interact with cementitious and lime-based matrices. Recent advancements include using cementitious suspensions to impregnate CF multifilament roving, which helps create rigid composite rods for reinforcing fine-grained cementitious composites [11]. The unpredictable crack propagation behavior in **carbon fiber reinforced plastic (CFRP)** composites poses a challenge to the safety and reliability of structures, as the material's performance is heavily affected by its crack propagation behavior [13]. Metallic materials with homogeneous structures have long faced the challenge of overcoming the typical trade-off between strength and ductility.[14].Hard and brittle materials,

such as silicon carbide, single-crystal silicon, and ceramic-based composites, exhibit outstanding mechanical, physical, and chemical properties. These materials find diverse applications in aerospace, aviation, microelectronics, medicine, and other industries.[15].Increasing both the compressive strength of concrete and the interface roughness improves load-carrying capacity, crack initiation toughness, and resistance to crack extension. Additionally, higher concrete compressive strength enhances the moment capacity of the interface.[16]

Fracture mechanics is a branch of engineering that uses mathematical models to determine when a part with a crack or defect will break, particularly under extreme stress. This technique is critical for ensuring that engineering parts are strong enough to perform safely. **Effectively designing gradient structures** allows precise control over material properties, optimizing the balance between strength and toughness, and improving material safety during use [1]. Fracture toughness, which measures a material's ability to resist crack propagation, is a key factor in assessing material safety in service [1]. Understanding these properties is vital for evaluating materials' performance under real-world conditions.

Specimen-size-independent **fatigue crack growth** data is essential for analyzing fatigue failures from a theoretical and practical perspective. This data reflects a material's true resistance to fatigue crack growth, and using it is crucial for accurate structural modeling, especially when mechanical structures vary in size. Within the current fracture mechanics framework, **fatigue crack growth (FCG)** correlations, based on the ΔK approach, are often assumed to be unaffected by specimen size or shape [2]. However, the possibility of size dependence in crack growth is highlighted by geometric correction factors (GCFs), which are used in calculating the stress intensity factor (K). For instance, **IT** has shown through numerical analysis that the stress intensity factor depends on the specimens' H/W ratio (height-to-width ratio) [2]. Material cracks cause damage, reducing mechanical properties like the axial modulus

under tensile loading and the shear modulus under shear loading, depending on crack orientation.

The use of **fiber- and textile-reinforced cementitious composites** has surged in recent decades due to their excellent mechanical properties, high shape flexibility, and relatively good durability. These composites are particularly valuable in both new structures and the reinforcement of existing ones. Thin-walled structures made from these composites provide effective protection against dynamic and extreme loading conditions [11]. Carbon fibers (CF) are widely used for multifilament rovings due to their excellent performance-to-cost ratio and resistance to aggressive environments. However, their **brittleness** and surface defects present challenges, especially under pull-out forces or when embedded in matrices containing sharp particles. To enhance the fibers' stability and longevity, polymer sizing and yarn impregnation techniques are employed [11].

Polymeric-impregnating agents, such as **epoxy, polyacrylate, and styrene-butadiene**, are often used to improve the mechanical properties of textile-reinforced composites (TRC). However, they can weaken the bond between fibers and substrates and reduce fire resistance, particularly at high temperatures. To address these issues, **inorganic impregnating agents**, like amorphous silica, have been developed to interact with cementitious and lime-based matrices. Recent advancements include using cementitious suspensions to impregnate CF multifilament roving, which helps create rigid composite rods for reinforcing fine-grained cementitious composites [11]. The unpredictable crack propagation behavior in **carbon fiber reinforced plastic (CFRP)** composites poses a challenge to the safety and reliability of structures, as the material's performance is heavily affected by its crack propagation behavior[13].

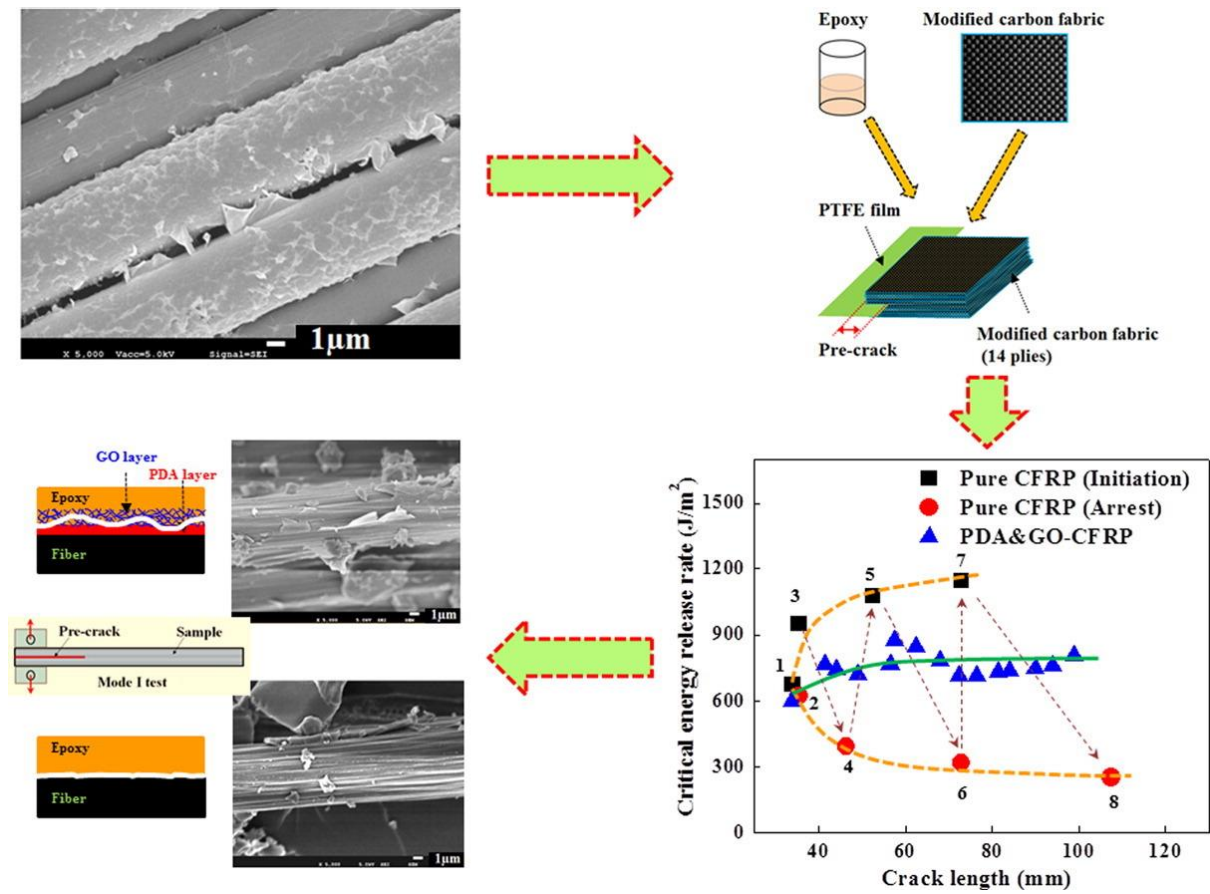


Figure 1 The different Crack behaviour of carbon fiber and modified composite carbon fiber

Note: derived from[13] [17]

Determining material fracture toughness is essential for fracture mechanics applications in design, material selection, and fitness-for-service assessments of load-bearing components. According to **ASTM E1820**, fracture toughness for materials exhibiting R-curve behavior can be determined using two methods: the multi-specimen methodology and the single-specimen methodology [18]. Both approaches require analysis of the specimen's fracture surface to determine physical crack extension (Δa), with the multi-specimen method quantifying crack extension for each specimen, while the single-specimen method validates crack monitoring results [18]. The **load increase** and subsequent load drop during **DCB** and **ENF tests** reveal the presence of unstable crack propagation, influenced by the varying material properties of the woven carbon fabric laminates used as substrates [19]

Cracks and imperfections in materials must be considered in engineering applications, as they can significantly affect the safety and performance of structures [20]. Given the extensive use of composites across various industries, predicting failure through suitable fracture criteria and models is crucial [20]. When cracks approach interfaces in a structure, the **fracture patterns** become complex. The presence of weak interfaces disrupts the defined crack paths, leading to unclear failure modes. Several factors, such as the strength and toughness of each phase, geometric properties, and loading conditions, influence the fracture pattern [21]. Additionally, the **strength** or **toughness of interfaces** significantly affects the final fracture pattern observed in layered or grain-based structures [21]. Cracks in concrete structures often propagate curvilinearly due to shear forces acting on the crack surface, necessitating remeshing for accurate **numerical analysis** [21].

Finally, **fracture mechanics in functionally graded materials (FGMs)** has gained attention, as failure in FGMs is more complex due to the non-uniform energy required for crack propagation. Early studies primarily focused on static linear elastic deformation of FGM materials [22]. Cracks and defects are inevitable during the production of structural elements, and understanding how these imperfections affect material behavior is essential for ensuring safety and functionality [23]. To predict crack paths and growth rates under **creep-fatigue conditions**, a creep-fatigue crack growth indicator parameter (CFCGIP) has been developed [1]. **Brittle fracture**, characterized by cleavage cracks and minimal energy absorption, is a critical factor in fracture mechanics [24]. Using **linear elastic fracture mechanics (LEFM)**, the stress intensity factor (SIF) can be used to characterize the stress field near the crack tip under different loading modes [23].

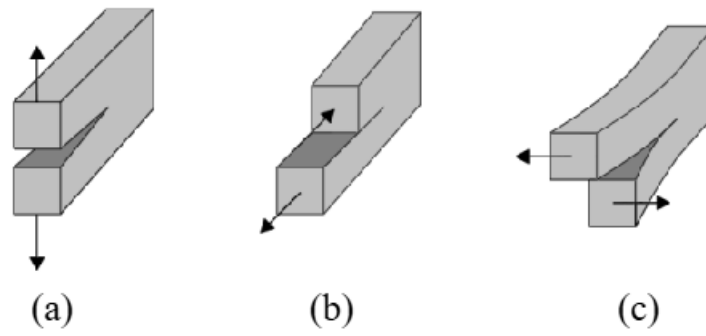


Figure 2 The three basic loading modes of crack are (a) Mode I (b) Mode II (c) Mode III

Note: derived from [23] [25]

Theory

The finite element method (FEM) is a useful tool in engineering for solving tough problems involving complex shapes, loads, or materials. It works by breaking a structure into small pieces called elements, which are connected at points called nodes. Software like Ansys and Abaqus makes FEM easier to use. In fracture mechanics, FEM helps study how cracks grow by analyzing key factors like stress intensity and energy release rates. This helps engineers understand crack behavior and design safer, stronger structures. The Finite Element Method (FEM) has undergone significant advancements in modeling fatigue crack propagation, particularly in composite materials. The effectiveness of FEM in predicting fatigue crack growth, leveraging adaptive mesh refinement techniques to enhance accuracy near crack tips. Their study highlighted the role of stress intensity factors (SIF) and energy release rates in understanding crack propagation under cyclic loading, providing a robust framework for analyzing fatigue behavior in complex structures[26] [27].

Mesh-separation-based approximation techniques have further advanced FEM's capabilities in modeling crack propagation. An adaptive mesh refinement strategy that improves the precision of stress and strain distribution calculations was introduced back in 2014. This approach allows for efficient tracking of crack growth paths while maintaining computational efficiency, making it suitable for problems involving intricate crack geometries and anisotropic material properties[28] [29].

Comparative studies have also shown the versatility of FEM in conjunction with other numerical methods. FEM was compared with the element-free Galerkin method and the extended finite element method (XFEM) in predicting SIF and energy release rates for anisotropic materials like cortical bone. The research demonstrated that FEM remains highly reliable for structural integrity analysis when incorporating anisotropic properties, validating its application in both biological and engineering materials[30] [31].

In brittle materials, the coupled criterion approach implemented using FEM has proven effective for simulating crack initiation and propagation. This method was explored to model brittle fracture, incorporating a balance between energy release rates and stress concentration criteria. This study emphasized the importance of using FEM to capture the interplay of mechanical and material factors influencing fracture processes, thereby providing deeper insights into crack behavior under various loading conditions[32].

Relative equations related to the subject:

- $\sigma = \varepsilon E$
- $\varepsilon = \frac{\delta}{L}$
- $P = \sigma A$
- $U = -\frac{\sigma}{2E} \pi a^2$

Model

We will be investigating the fracture behavior of a composite plate by analyzing internal cracks. These cracks reveal imperfections and flaws in the material, which can affect its performance. By studying the fracture configuration, we can understand how the plate responds to stress and its impact on structural integrity. The plate's dimensions remain the same, except for the thickness, which is varied at 0.3 mm, 0.5 mm, 1 mm, and 2 mm.

Simulation

In this simulation section we used Ansys to analyze the fracture behavior of a composite plate. We focused on understanding how cracks develop and grow under 10 MPa pressure fixated on one face. The material chosen is Carbon Fiber composite which offers a great tensile strength and has a modulus of elasticity between 200-500 GPa making it an ideal material for different applications[34] [35]. The figure 3 below shows the material properties.

Properties of Outline Row 3: Carbon Fiber (230 GPa)				
	A	B	C	D E
1	Property	Value	Unit	 
2	 Density	1800	kg m ⁻³	 
3	 Orthotropic Elasticity			
4	Young's Modulus X direction	230	GPa	
5	Young's Modulus Y direction	23	GPa	
6	Young's Modulus Z direction	23	GPa	
7	Poisson's Ratio XY	0.2		
8	Poisson's Ratio YZ	0.4		
9	Poisson's Ratio XZ	0.2		
10	Shear Modulus XY	9	GPa	
11	Shear Modulus YZ	8.2143	GPa	
12	Shear Modulus XZ	9	GPa	

Figure 3 Material Properties

The aim of this section we analyzed how thickness of the chosen material affects the crack propagation and stress intensity factor. The fracture toughness used in the analysis is $49\text{Mpa}\sqrt{\text{m}}$ found from [36] [37]. Initial conditions we kept a fixed support on one face and applied pressure on the opposite face. The pressure applied was kept constant throughout the analysis when involving different thickness and the amount was 10Mpa.

Step by step process of simulation procedure:

In ANSYS, the Static Structural module was added to the project schematic, the appropriate material was assigned from the Engineering Data, and a 2D sketch was created in SpaceClaim to initiate the model design process. We kept the 2D dimensions constant throughout the analysis and only varying the thickness.

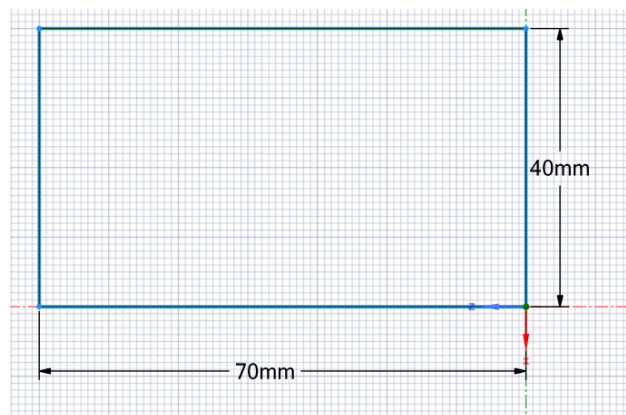


Figure 4 2D Sketch of the Composite Plate

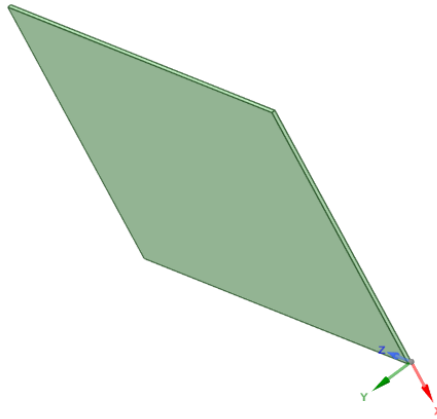


Figure 5 Extruded section of the drawing

Conditions were to keep one end of the plate fixed using fixed point tool found in analysis settings and applied pressure on the opposite end.

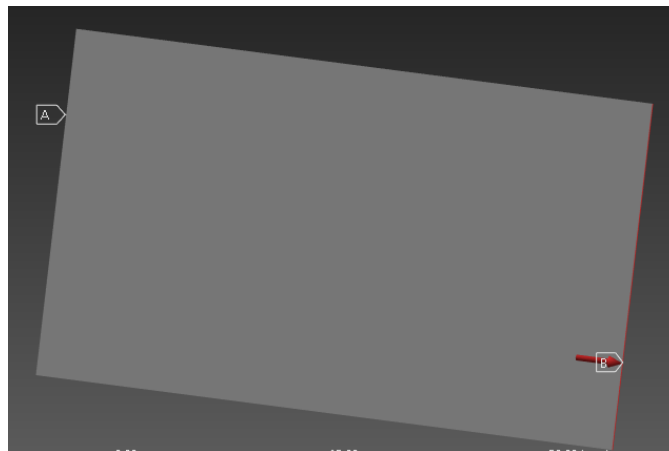


Figure 6 Image showing the conditions considered during the analysis

Next step was to insert and new coordinate system for the semi-elliptical crack.

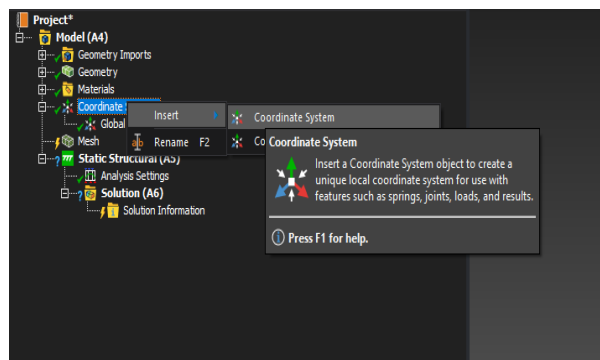


Figure 7 Figure showing the assignment of the coordinate system

We then went on to insert fracture and inserted semi-elliptical crack then changed the cracks orientation to ensure that it is mode-1 dominant. Tetrahedrons mesh was utilized for the analysis including the crack.

Scope	
Source	Analytical Crack
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Crack ID	33
Coordinate System	Coordinate System
Align with Face Normal	Yes
Project to Nearest Surface	Yes
Crack Shape	Semi-Elliptical
Major Radius	0.4 mm
Minor Radius	0.2 mm
Mesh Method	Tetrahedrons
Largest Contour Radius	0.25 mm
Growth Rate	Default (1.2)
Front Element Size	Default (2.5176e-002 mm)
Mesh Contours	6

Figure 8 Semi-elliptical crack details

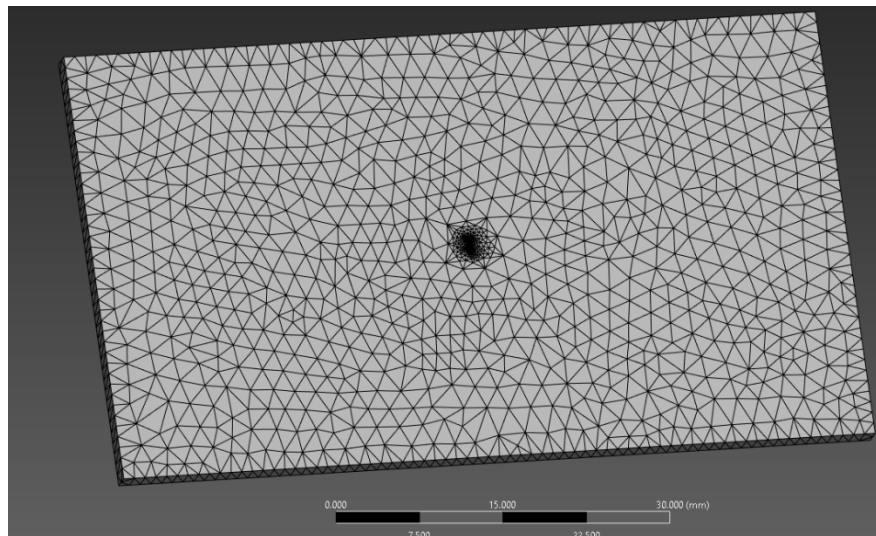


Figure 9 Tetrahedron mesh on the plate

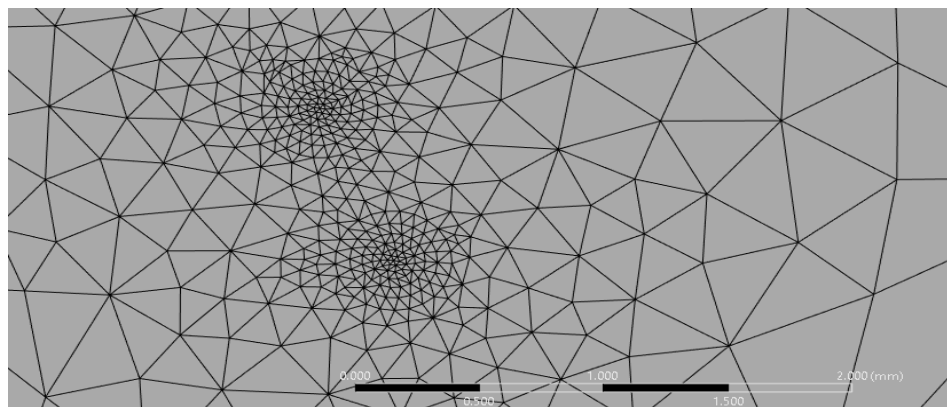


Figure 10 Tetrahedron mesh on the semi-elliptical crack

Before solving, we ensured that J-integral was kept off since the system cannot solve for the two at the same time and our main focus was on stress intensity factor (SIFS). The details can be found on analysis settings. Figure 11 shows the outline tree.

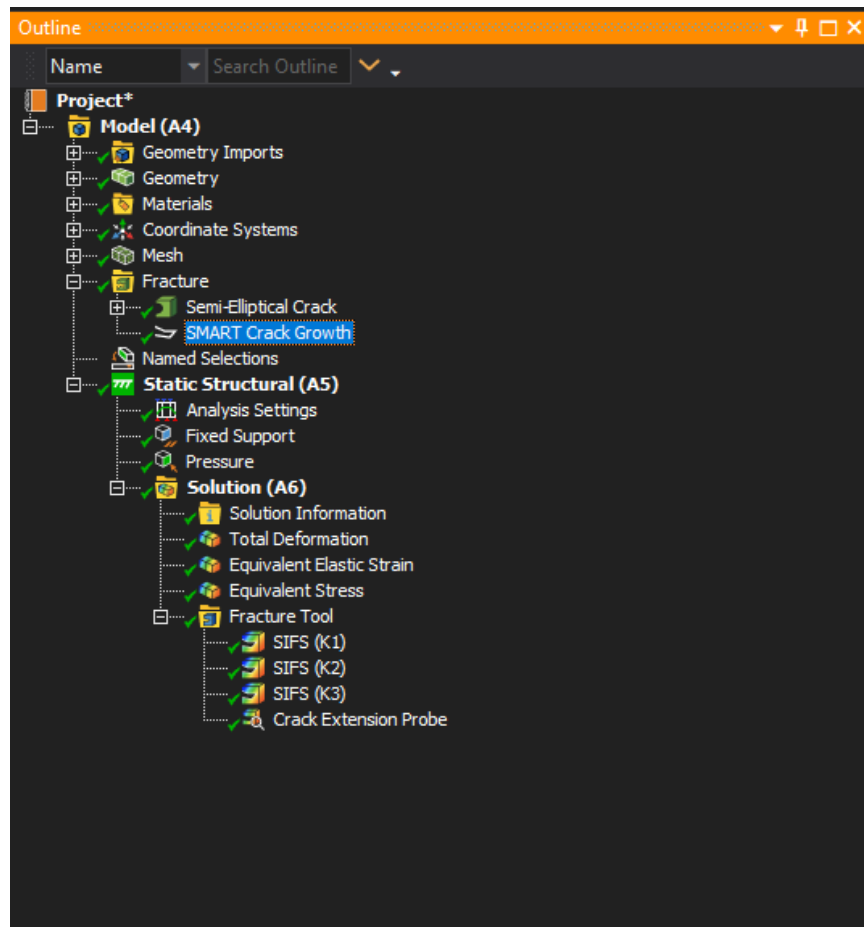


Figure 11 Outline tree

Solution

As shown fig.2, the three basic modes are: opening (mode 1), shear (mode 2), tearing (mode 3). Mode I corresponds to normal separation of the crack faces under the action of tensile stresses, which is by far the most widely encountered in practice. The difference between Mode II and Mode III is that the shearing action in the former case is normal to the crack front in the plane of the crack whereas the shearing action in Mode III is parallel to the crack front. A cracked body in reality can be loaded in any one of these three, or a combination of these three modes.[38]

Stress Intensity factor of different modes at different thickness

The **Figure 12-23** illustrate the variation of stress intensity factors K1, K2, and K3 for a material with thicknesses of 0.3 mm, 0.5 mm, 1 mm, and 2 mm, providing insights into crack behavior under different loading conditions. K1 (Mode-I) represents the tensile opening mode, which is typically dominant and critical for crack initiation, particularly in thinner materials. K2 (Mode-II) captures in-plane shear stresses, relevant in mixed-mode loading scenarios, while K3 (Mode-III) reflects out-of-plane shear or tearing effects, which become more significant with increasing thickness. By analyzing these factors across varying thicknesses, the graphs highlight how material geometry influences stress distribution and fracture mechanisms, aiding in the design of more robust structures and materials.

Graph of 0.3 mm thickness:

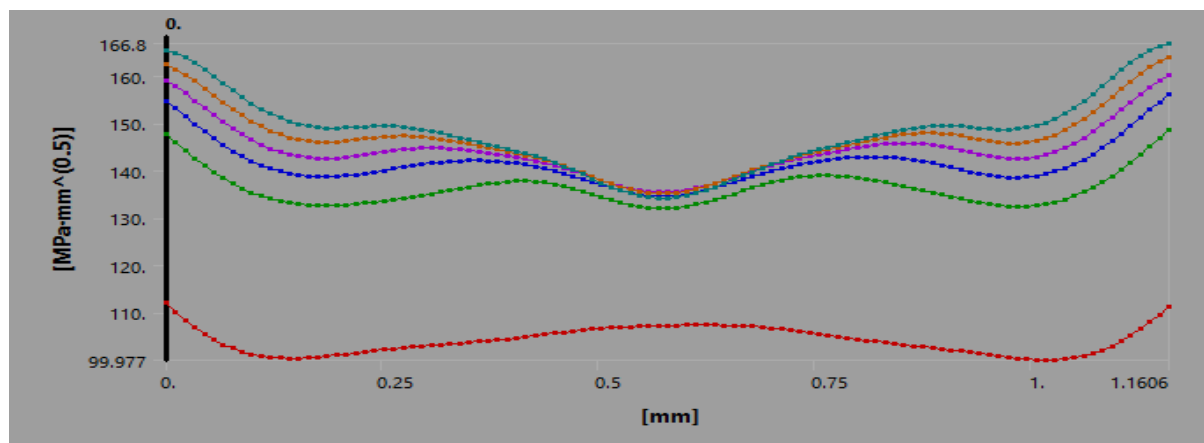


Figure 12 K1 value for 0.3 mm thickness

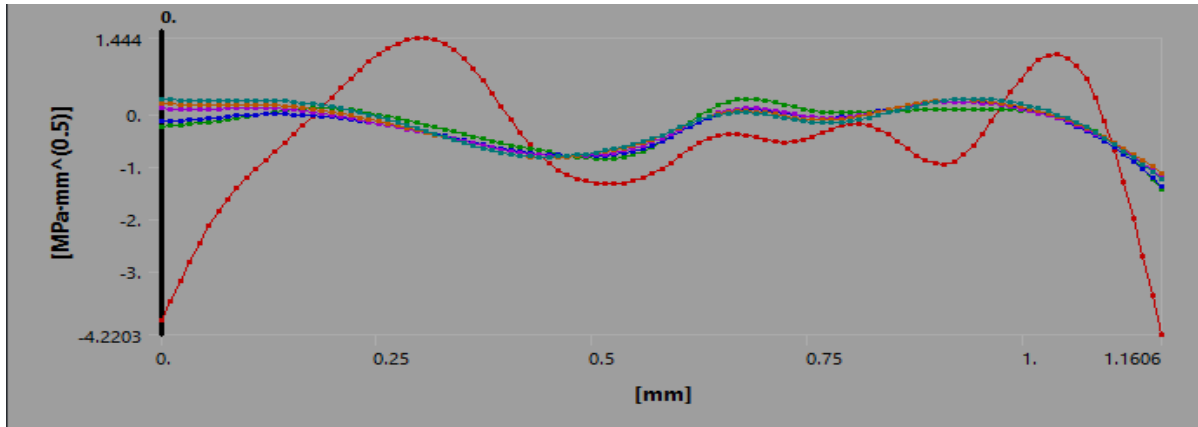


Figure 13 K2 value for 0.3 mm thickness

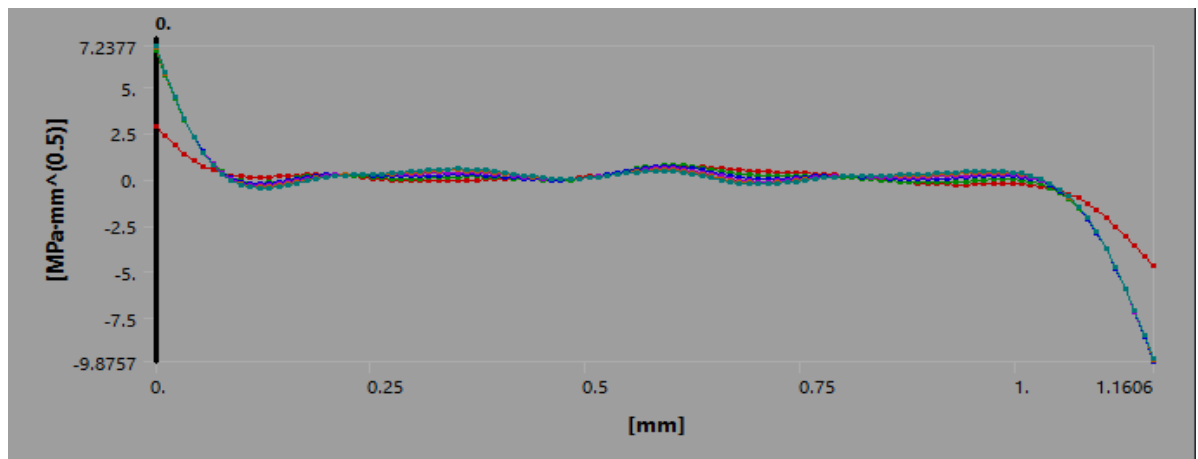


Figure 14 K3 value for 0.3 mm thickness

Graph of 0.5 mm thickness:

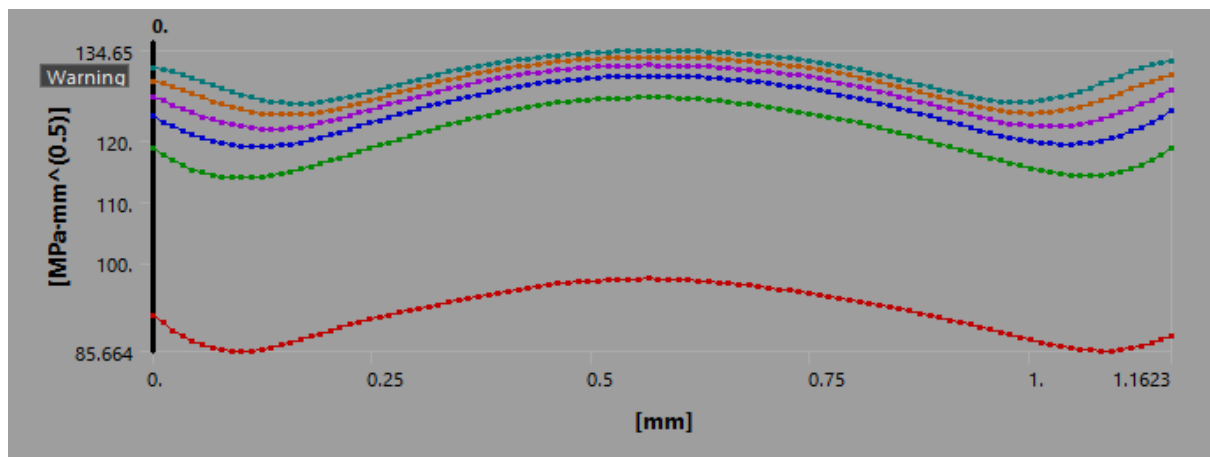


Figure 15 K1 value for 0.5 mm thickness

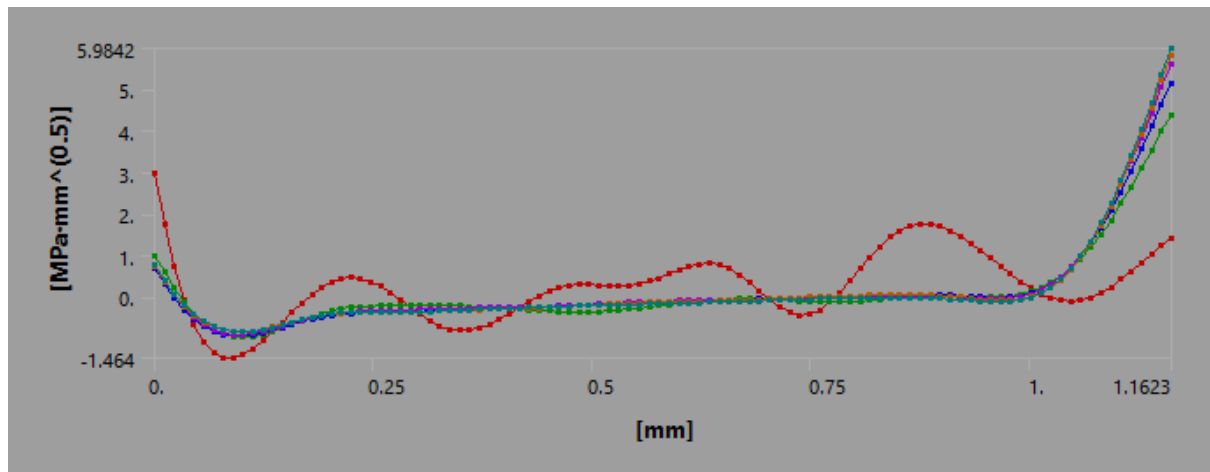


Figure 16 K2 value for 0.5 mm thickness

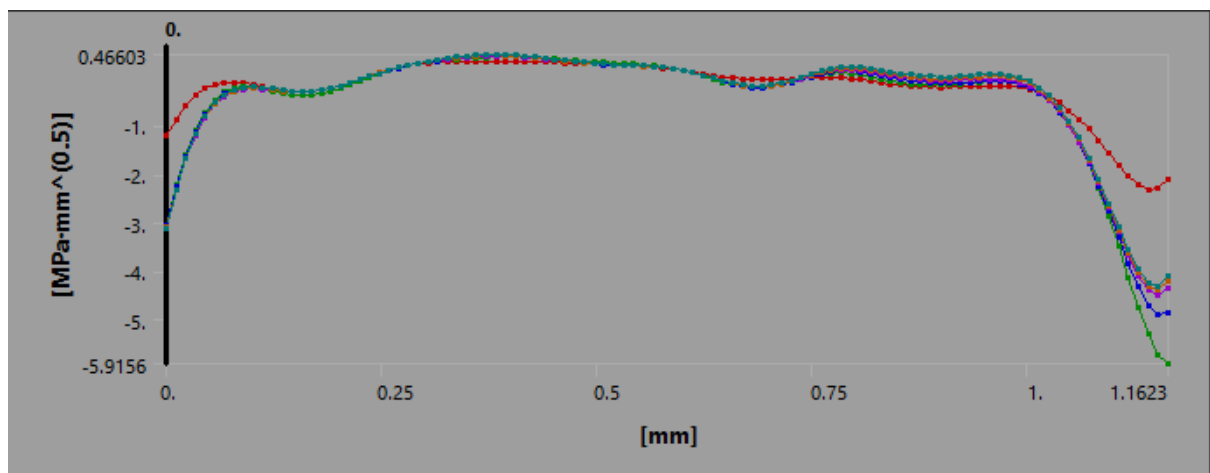


Figure 17 K3 value for 0.5 mm thickness

Graph of 1 mm thickness:

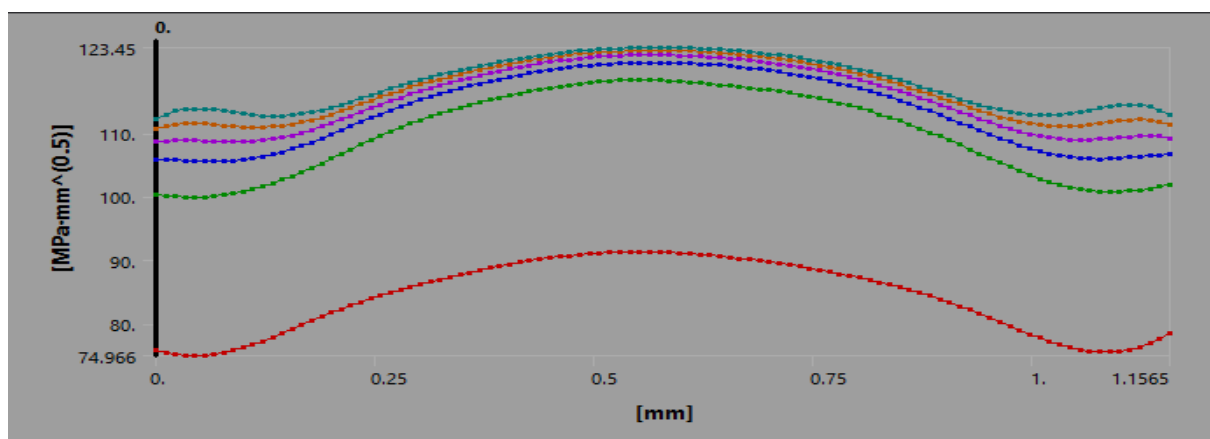


Figure 18 K1 value for 1 mm thickness

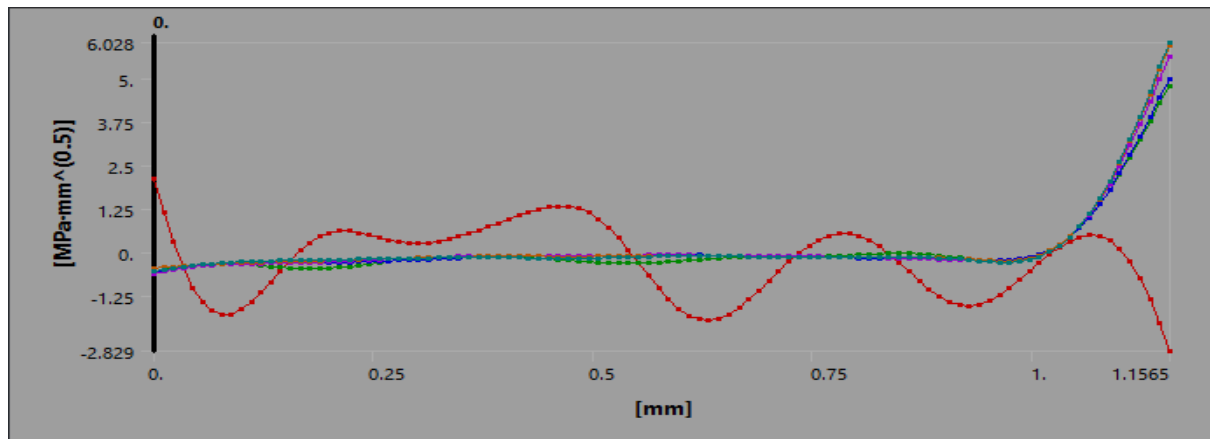


Figure 19 K2 value for 1 mm thickness

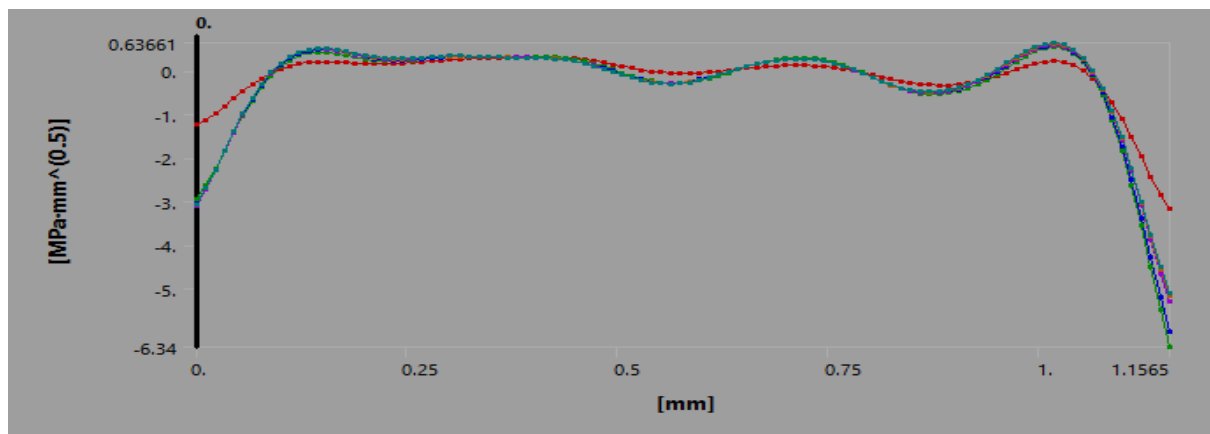


Figure 20 K3 value for 1 mm thickness

Graph of 2 mm thickness:

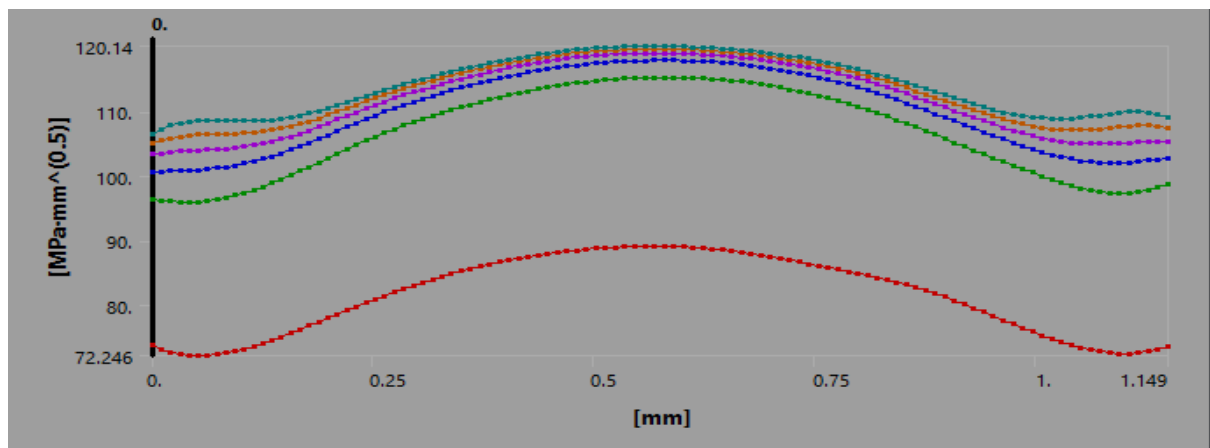


Figure 21 K1 value for 2 mm thickness

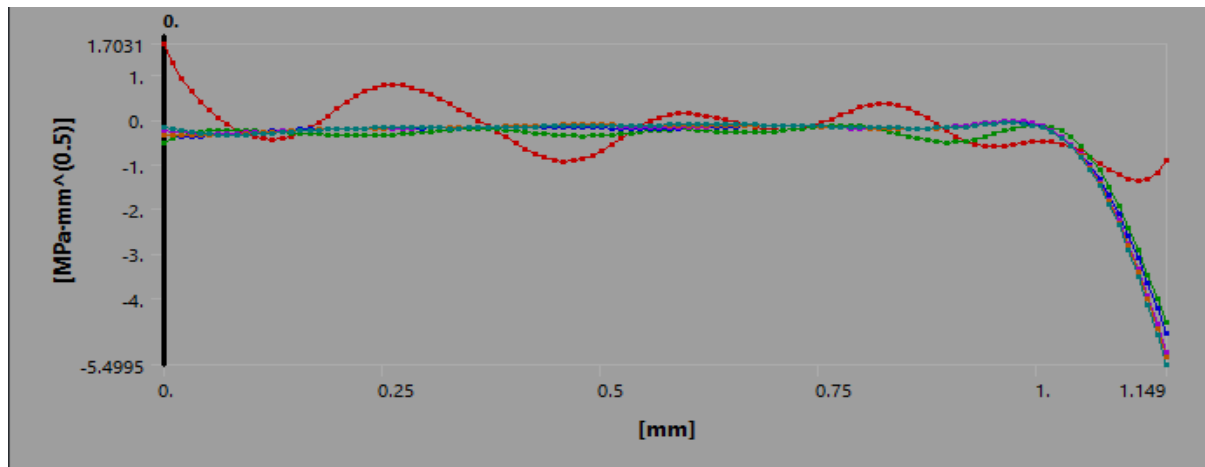


Figure 22 K2 value for 2 mm thickness

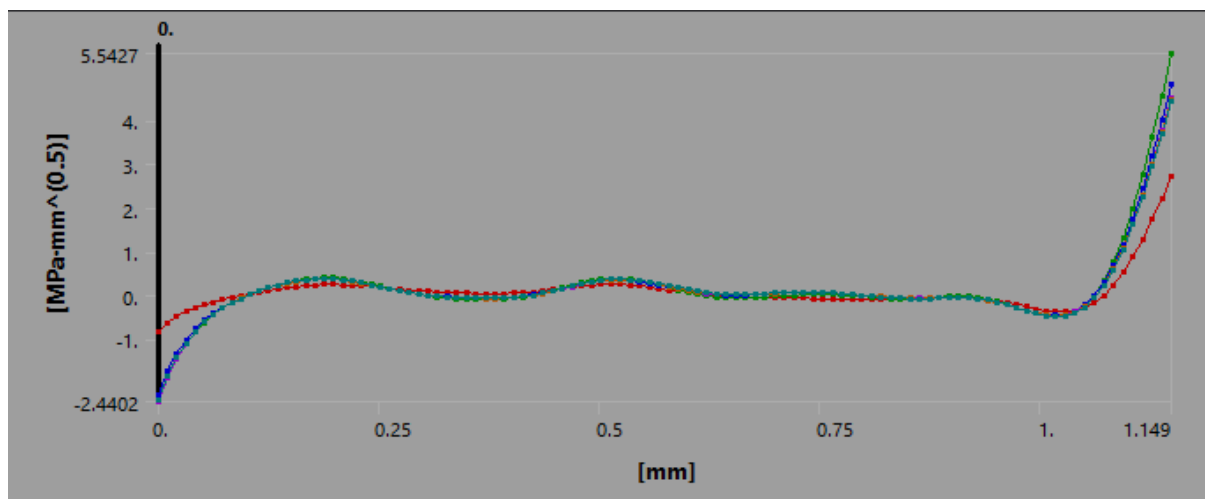


Figure 23 K3 value for 2 mm thickness

Equivalent Stress along the plate:

Equivalent stress as known also von mises stress, von Mises simply enables engineers to understand the performance of a complexly loaded part with a simple uniaxial tensile test. The equivalent stresses are displayed in **Figure 24-30**. These can be interpreted those high stresses located inside the fracture, with different maximum value for each different thickness.

Equivalent Stress along the plate at 0.3 mm thickness:

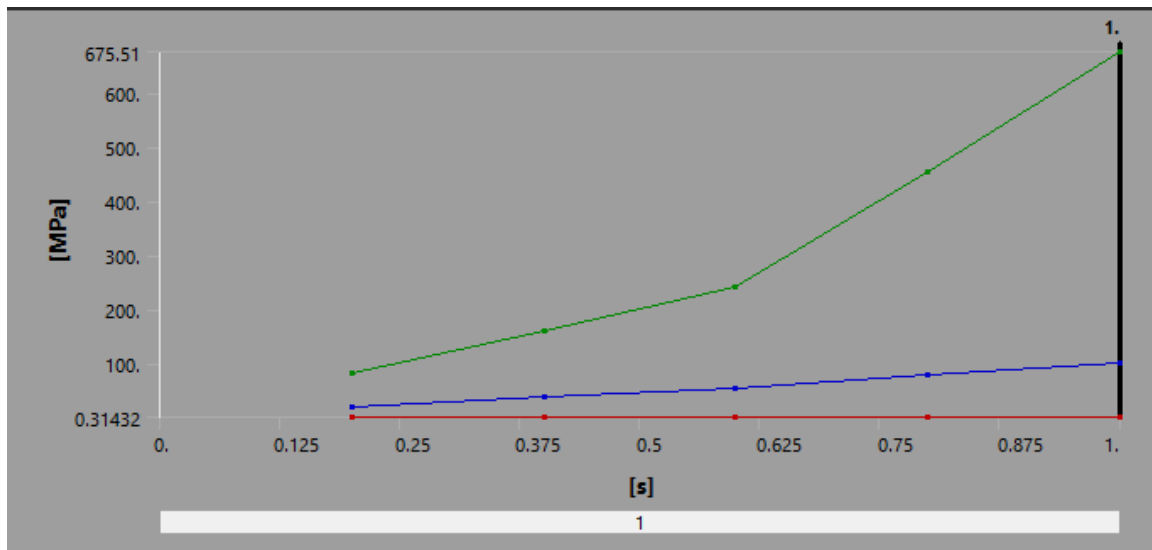


Figure 24 Stresses on X,Y,Z with time for 0.3 mm thickness

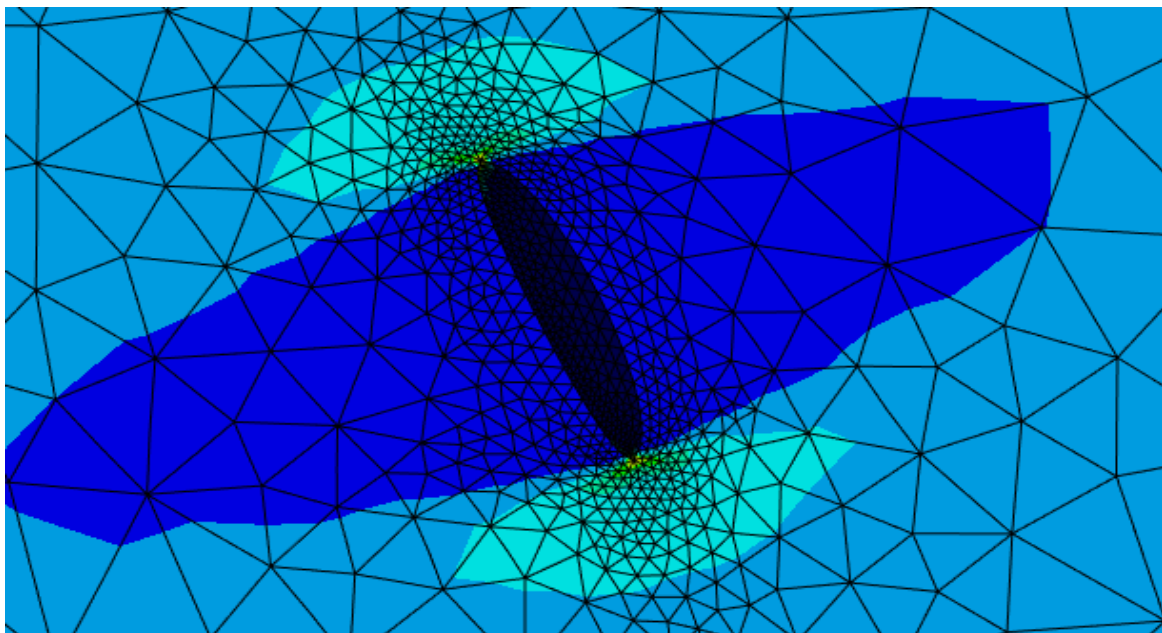


Figure 25 Schematic stress representation of 0.3 mm thickness on uniaxial load

Equivalent Stress along the plate at 0.5 mm thickness:

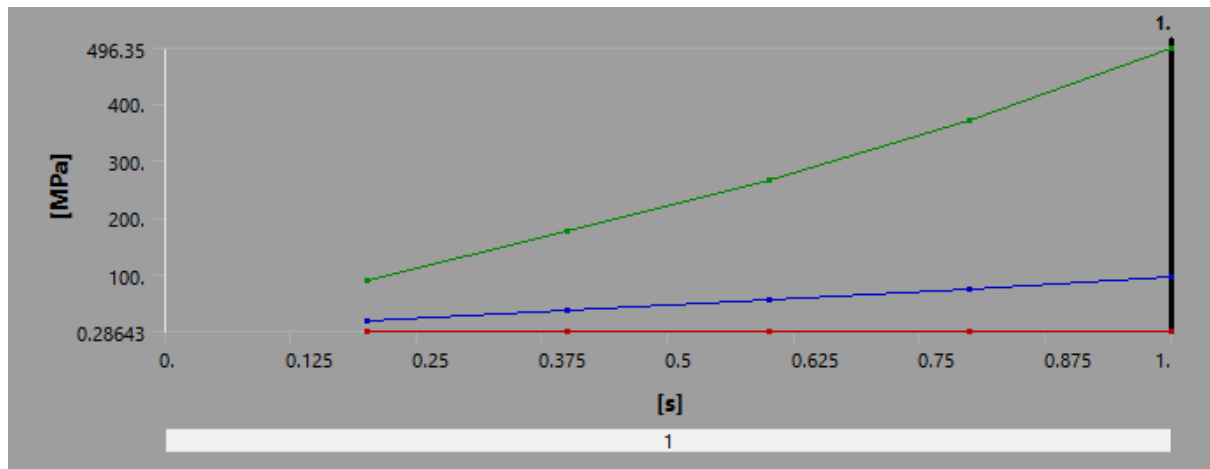


Figure 26 Stresses on X, Y, Z with time for 0.5 mm thickness

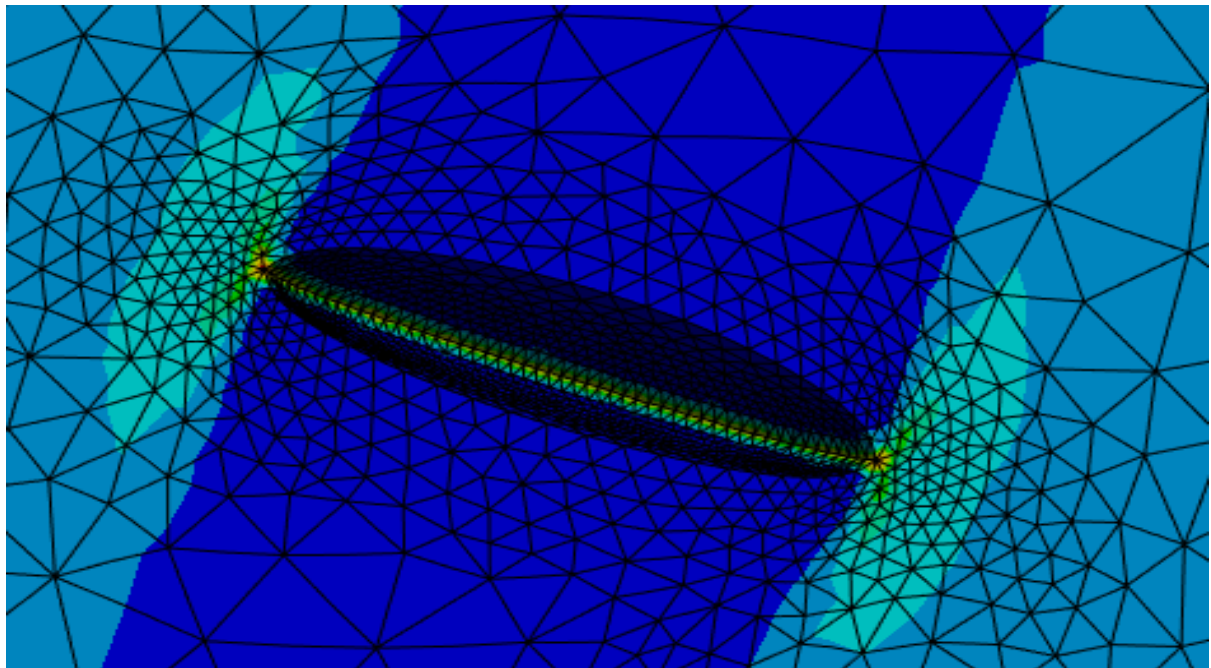


Figure 27 Schematic stress representation of 0.5 mm thickness on uniaxial load

Equivalent Stress along the plate at 1 mm thickness:

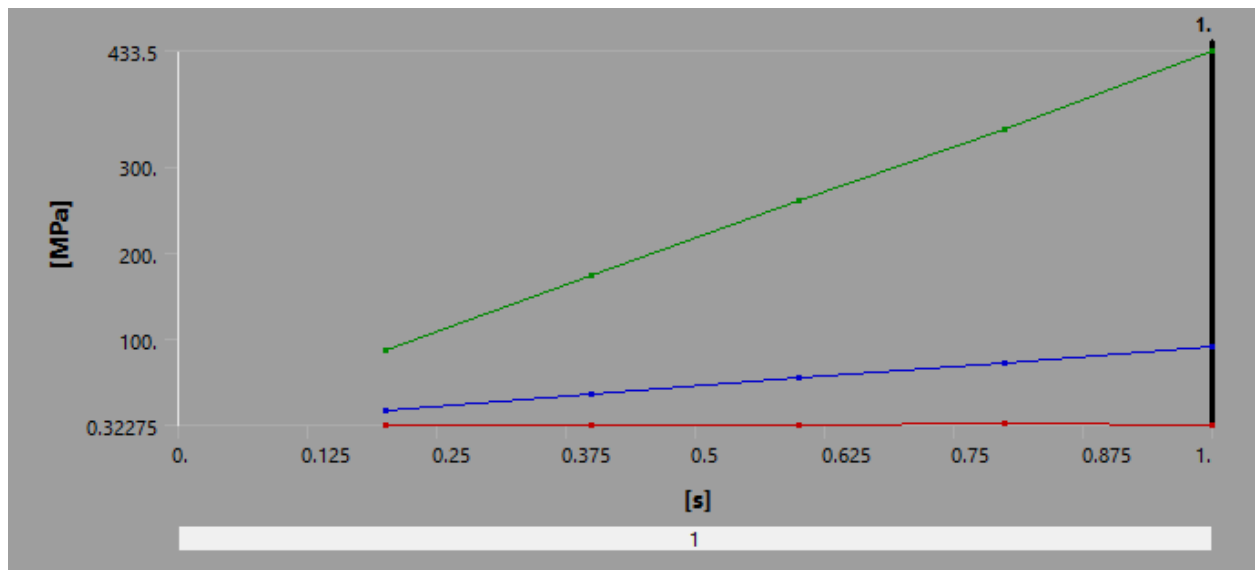


Figure 28 Stresses on X,Y,Z with time for 1 mm thickness

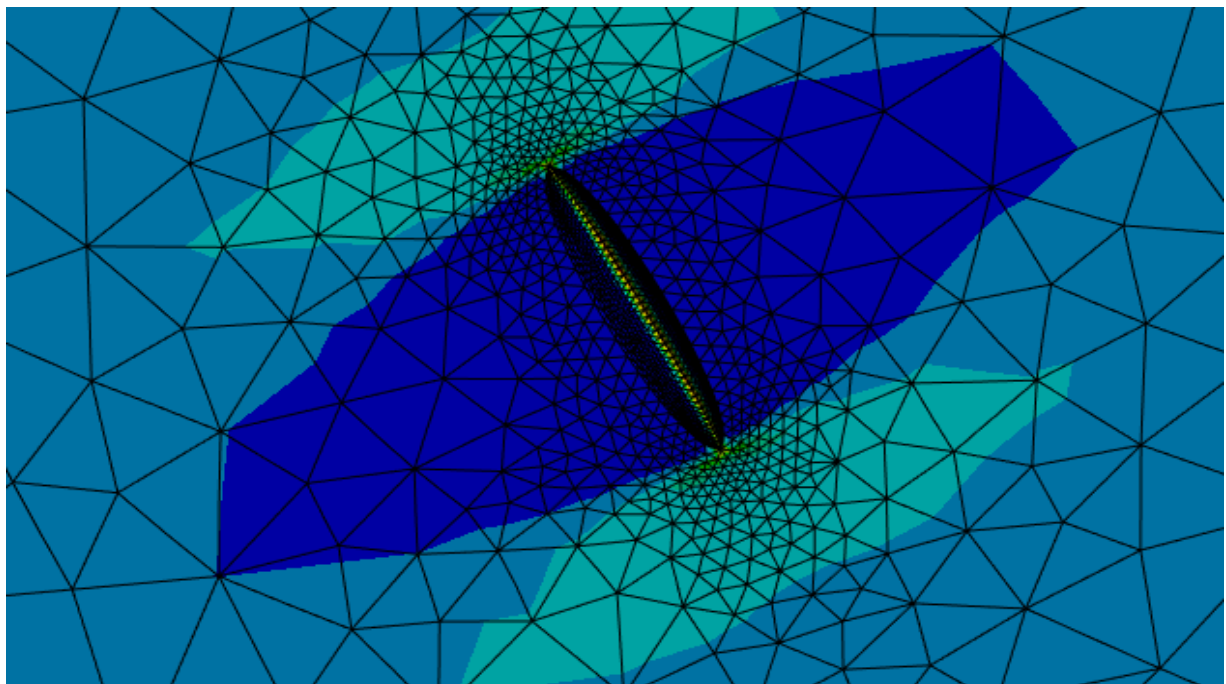


Figure 29 Schematic stress representation of 1 mm thickness on uniaxial load

Equivalent Stress along the plate at 2 mm thickness:

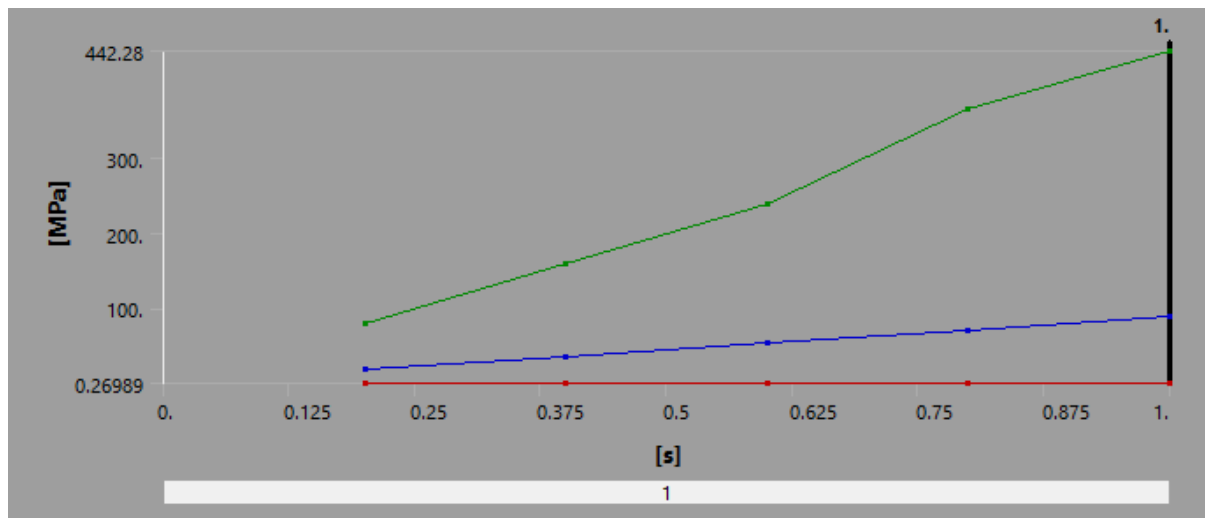


Figure 30 Stresses on X,Y,Z with time for 2 mm thickness

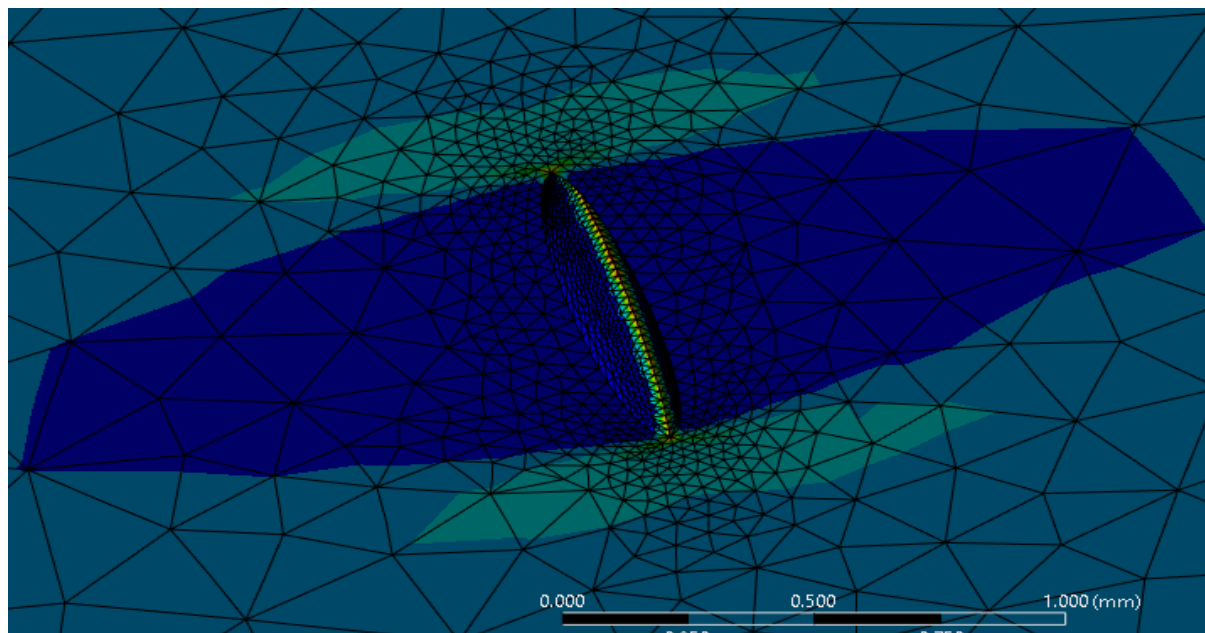


Figure 31 Schematic stress representation of 2 mm thickness on uniaxial load

The figures from 24-31 shows that, the maximum stress decrease while thickness increase but at 2 mm thickness the value of the maximum stress higher than 1 mm thickness. The reason behind of it basically anisotropy. Composite materials also known as anisotropic material. Within each grain the behavior is anisotropic due to the crystal planes, and if a grain boundary is crossed, the orientation of these planes' changes. That shows to the reason behind of the why after 1mm maximum stress increase again.

Equivalent of strain in along plate

The equivalent elastic strain provides a unified measure of material deformation under elastic loading conditions, integrating normal and shear strains into a single scalar value. It is a key parameter in understanding material behavior under complex loading scenarios, particularly in regions near stress concentrations such as cracks. This measure helps evaluate the point at which a material transitions from elastic to plastic behavior, identified as the elastic limit on the stress-strain curve.

Figures 32–39 illustrate the strain distribution across various thicknesses, focusing on the crack region where strain concentration is most significant. The blue regions in the strain schematic represent areas of minimal deformation, indicating lower strain intensities or regions farther from the crack's influence. In contrast, the red or orange regions highlight zones of maximum strain concentration, typically located around the crack tip where the material experiences the highest deformation. These visualizations emphasize the critical role of strain localization in assessing material performance and potential failure. The study provides insights into strain evolution and its implications for material design and durability.

Equivalent of strain along plate at 0.3 mm thickness:

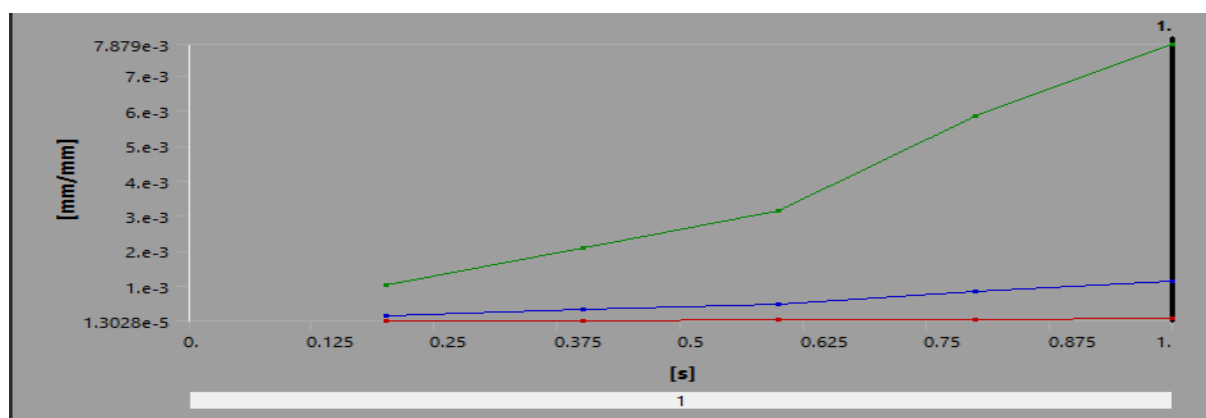


Figure 32 Strain graphs on X, Y, Z components for 0.3 mm thickness

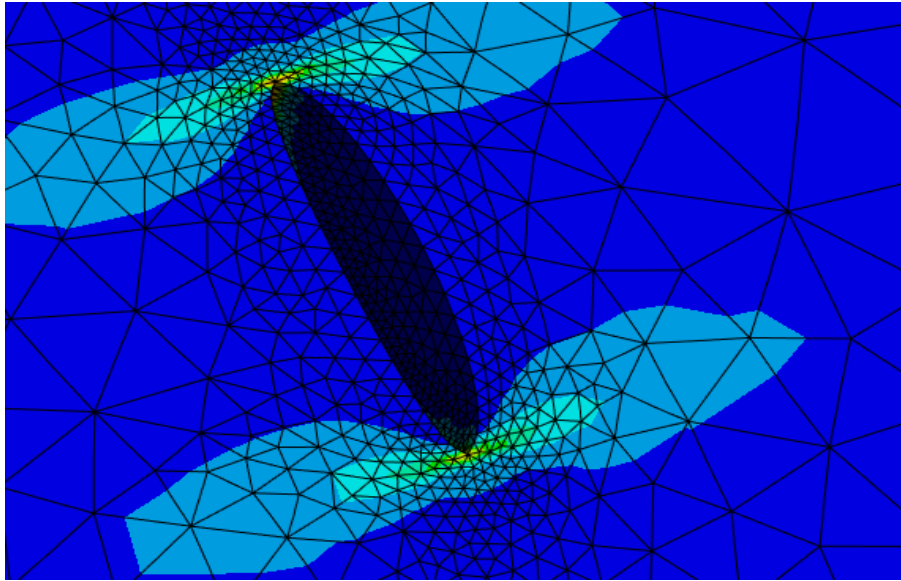


Figure 33 Schematic strain representation of 0.3 mm thickness

Equivalent of strain along plate at 0.5 mm thickness:

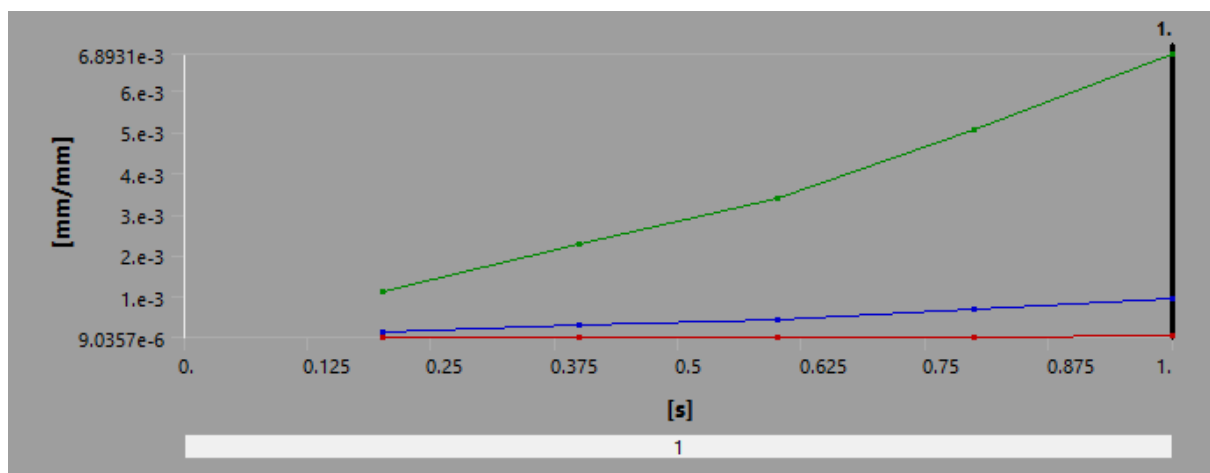


Figure 34 Strain graphs on X, Y, Z components for 0.5 mm thickness

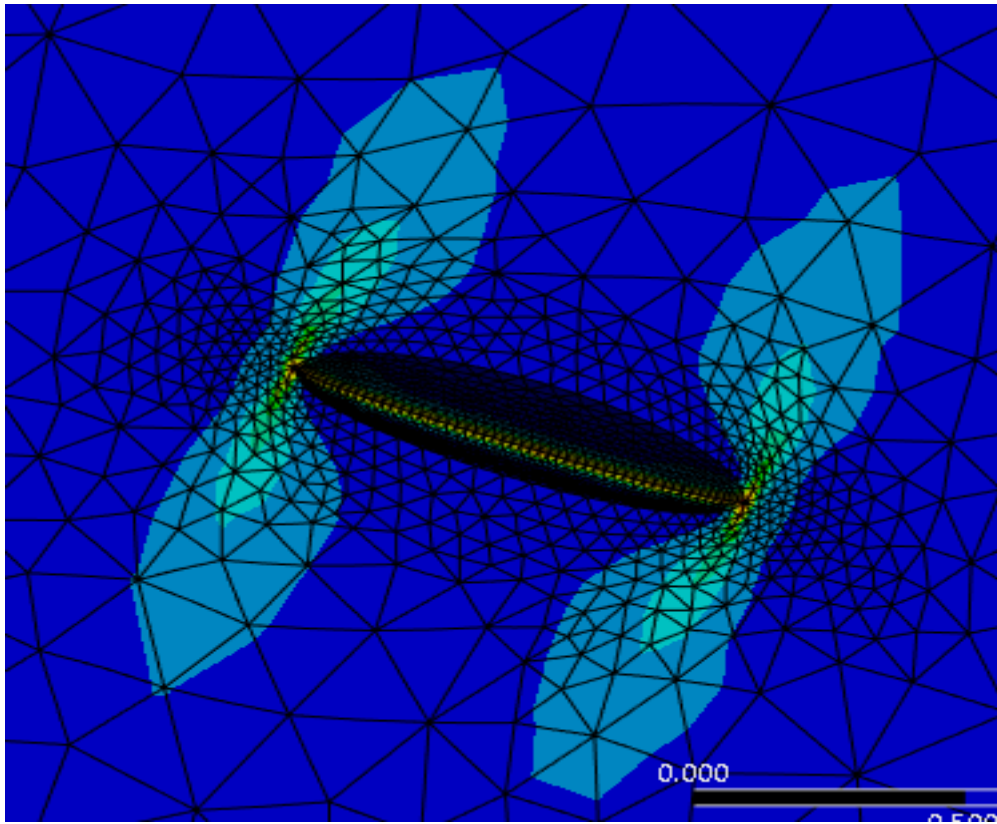


Figure 35 Schematic strain representation of 0.5 mm thickness

Equivalent of strain along plate at 1 mm thickness:

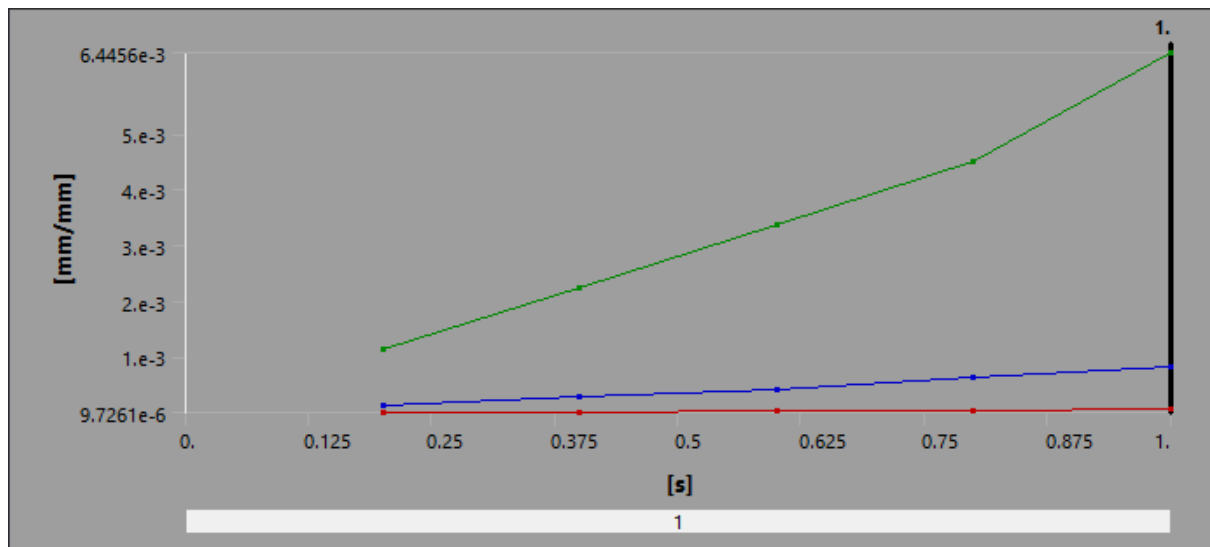


Figure 36 Strain graphs on X, Y, Z components for 1 mm thickness

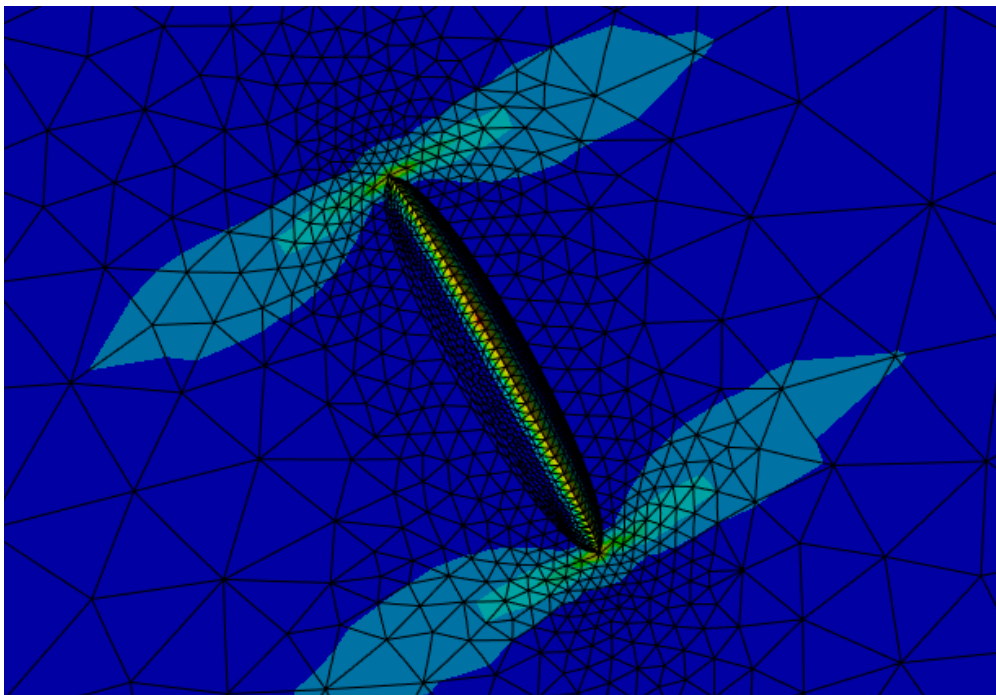


Figure 37 Schematic strain representation of 1 mm thickness

Equivalent of strain along plate at 2 mm thickness:

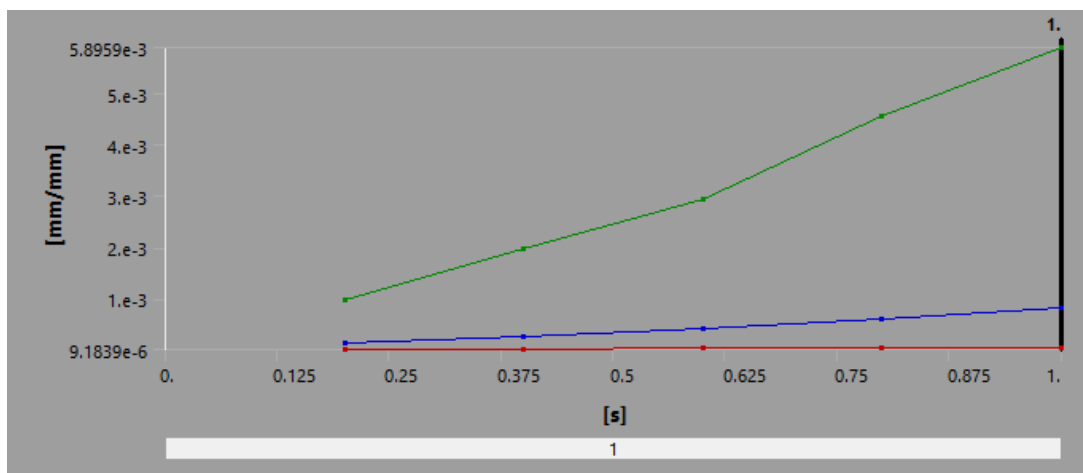


Figure 38 Strain graphs on X, Y, Z components for 2 mm thickness

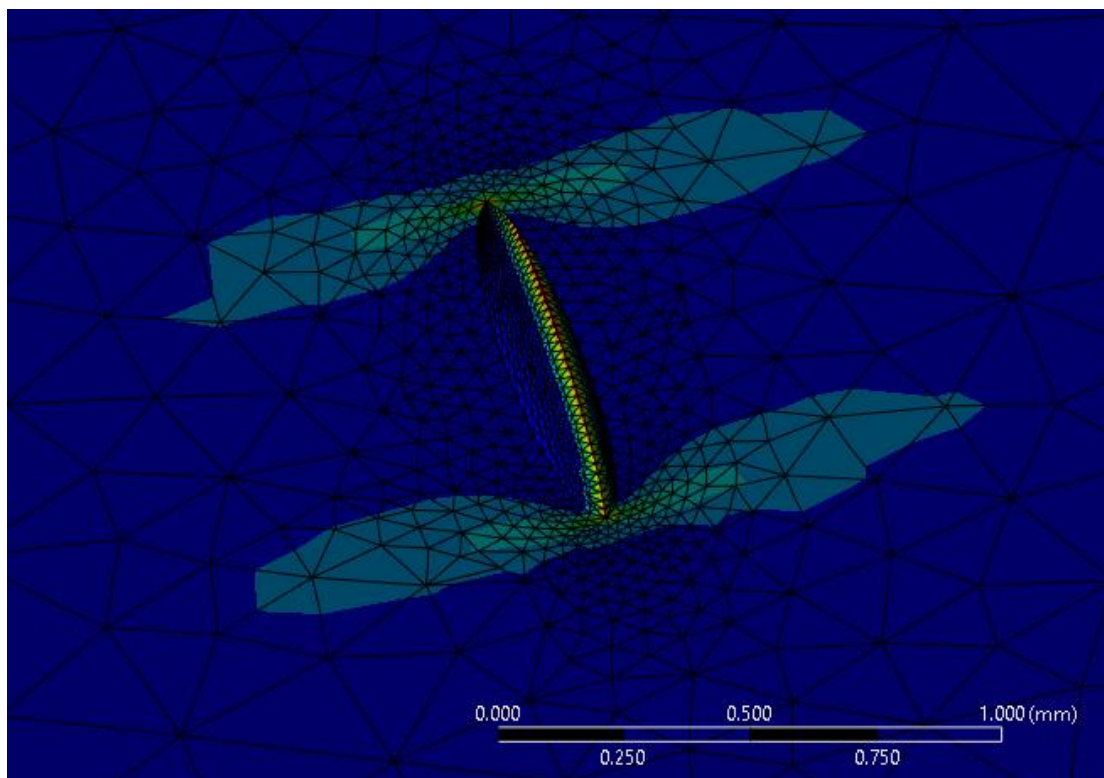


Figure 39 Schematic strain representation of 2 mm thickness

CRACK EXTENSION

Crack extension is a critical phenomenon in fatigue failure, where cracks propagate over time due to cyclic loading and stress concentration. Cracks may originate during manufacturing or develop under operational conditions, growing incrementally along planes perpendicular to the maximum tensile stresses. The progression of crack length significantly impacts the structural integrity and lifespan of materials, making its analysis essential for predicting failure and enhancing material design.

Figures 40–47 illustrate crack extension behavior for plates of varying thicknesses (0.5 mm, 1 mm, and 2 mm), with data obtained using a crack extension probe. This tool accurately tracks crack growth over time, enabling detailed analysis of the relationship between time and crack propagation. The results highlight how thinner plates tend to exhibit faster crack growth due to higher stress intensities near the crack tip. The accompanying tables summarize the crack extension at specific intervals, providing a clear quantitative overview. This study offers valuable insights into fatigue mechanisms and the influence of material thickness on crack progression, aiding in the design of more durable structures.

Crack extension for 0.3 mm thickness plate:

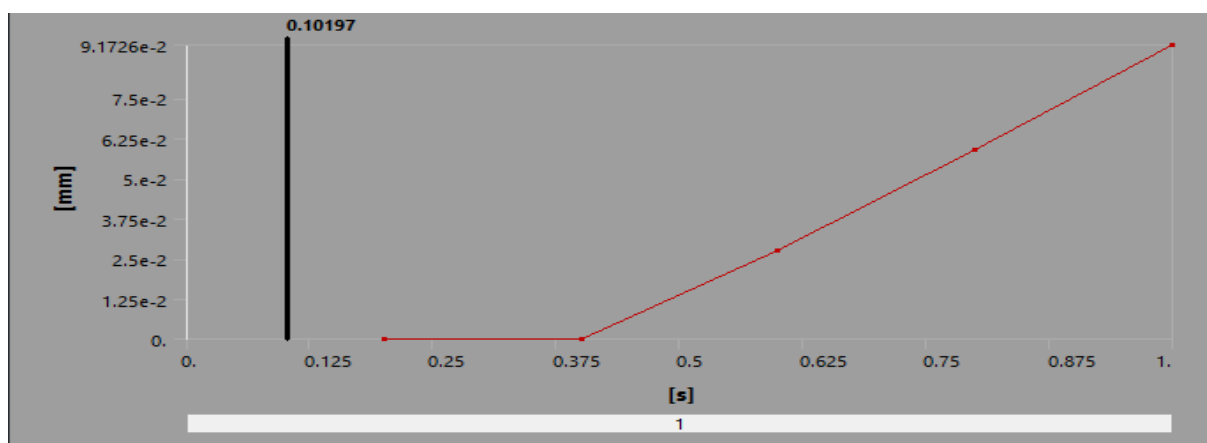


Figure 40 Crack extension graph with time for 0.3 mm thickness plate

	Time [s]	☒ Crack Extension Probe [mm]	Changed Mesh
1	0.2	0.	
2	0.4	0.	
3	0.6	2.7463e-002	
4	0.8	5.9085e-002	Yes
5	1.	9.1726e-002	Yes

Figure 41 Crack extension probe for every 0.2 seconds on 0.3 mm thickness plate

Crack extension for 0.5 mm thickness plate:

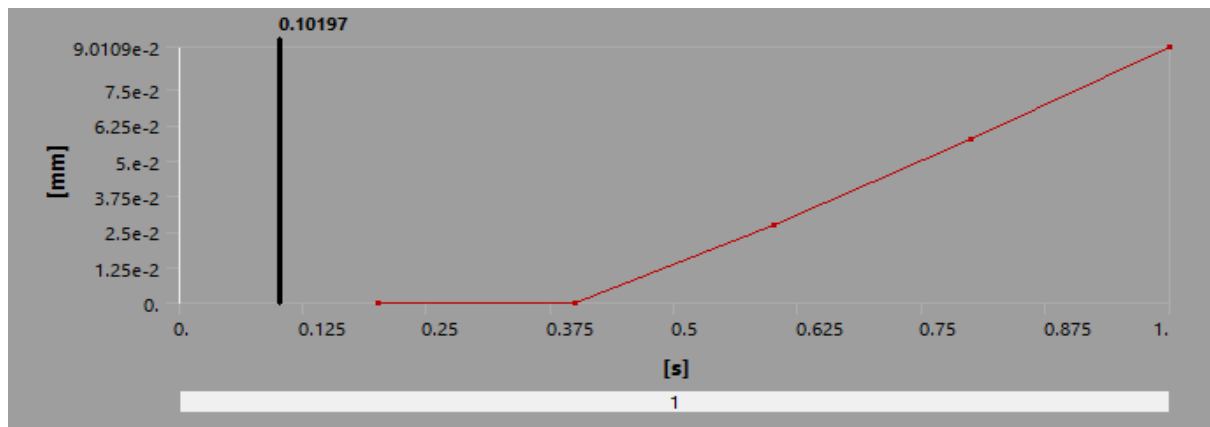


Figure 42 Crack extension graph with time for 0.5 mm thickness plate

	Time [s]	☒ Crack Extension Probe [mm]	Changed Mesh
1	0.2	0.	
2	0.4	0.	
3	0.6	2.7526e-002	
4	0.8	5.7898e-002	Yes
5	1.	9.0109e-002	Yes

Figure 43 Crack extension probe for every 0.2 seconds on 0.5 mm thickness plate

Crack extension for 1 mm thickness plate:_

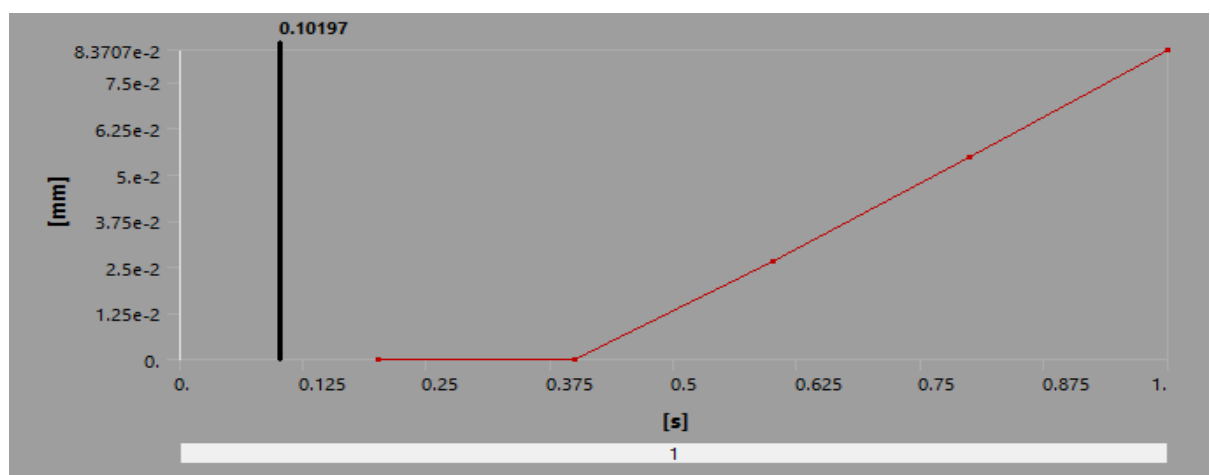


Figure 44 Crack extension graph with time for 1 mm thickness plate

	Time [s]	☒ Crack Extension Probe [mm]	Changed Mesh
1	0.2	0.	
2	0.4	0.	
3	0.6	2.652e-002	
4	0.8	5.4684e-002	Yes
5	1.	8.3707e-002	Yes

Figure 45 Crack extension probe for every 0.2 seconds on 1 mm thickness plate

Crack extension for 2 mm thickness plate:

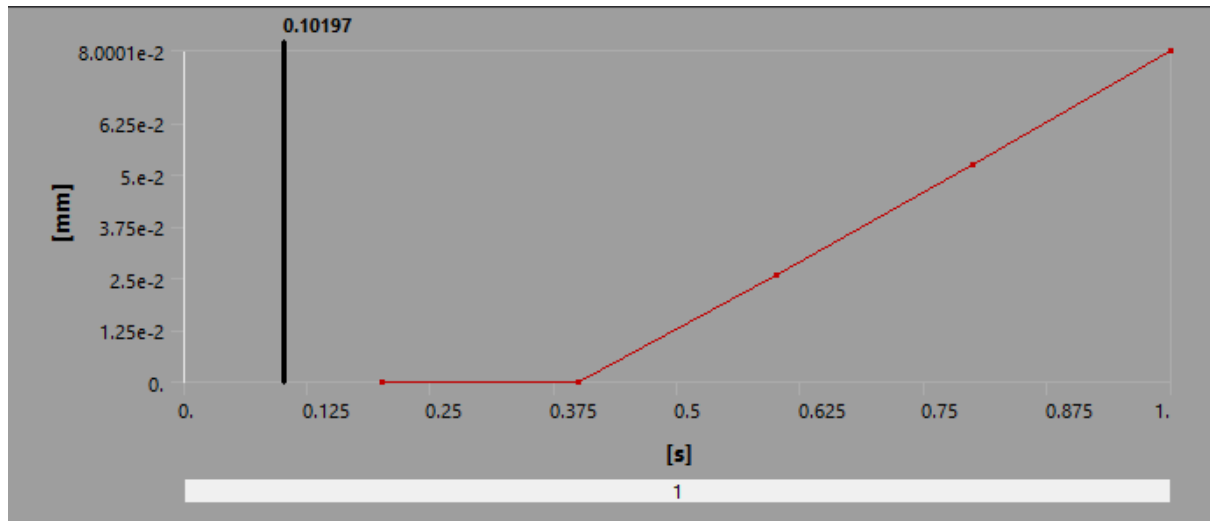


Figure 46 Crack extension graph with time for 2 mm thickness plate

	Time [s]	☒ Crack Extension Probe [mm]	Changed Mesh
1	0.2	0.	
2	0.4	0.	
3	0.6	2.5655e-002	
4	0.8	5.2469e-002	Yes
5	1.	8.0001e-002	Yes

Figure 47 Crack extension probe for every 0.2 seconds on 2 mm thickness plate

Table 1 Each thickness with their collected data.

Thickness (mm)	Fracture toughness (MPa)	Stress intensity factor (MPa.mm ^{0.5})	Max elongation (mm)	Maximum stress (MPa)	Maximum strain (mm/mm)
0.3	49	166.8	9.1726x10 ⁻²	675.51	7.879x10 ⁻³
0.5	49	134	9.0109x10 ⁻²	496.35	6.8931x10 ⁻³
1	49	123.45	8.3707x10 ⁻²	433.5	6.4456x10 ⁻³
2	49	120.14	8.001x10 ⁻²	442.28	5.8959x10 ⁻³

Discussion

The relationship between the SIFs(stress intensity factor),energy release rates and crack propagation paths are important for fracture behavior understanding. These notions enable the analysis of different behaviors in material thickness, depending on stress , as expressed by the graphs and simulations presented .

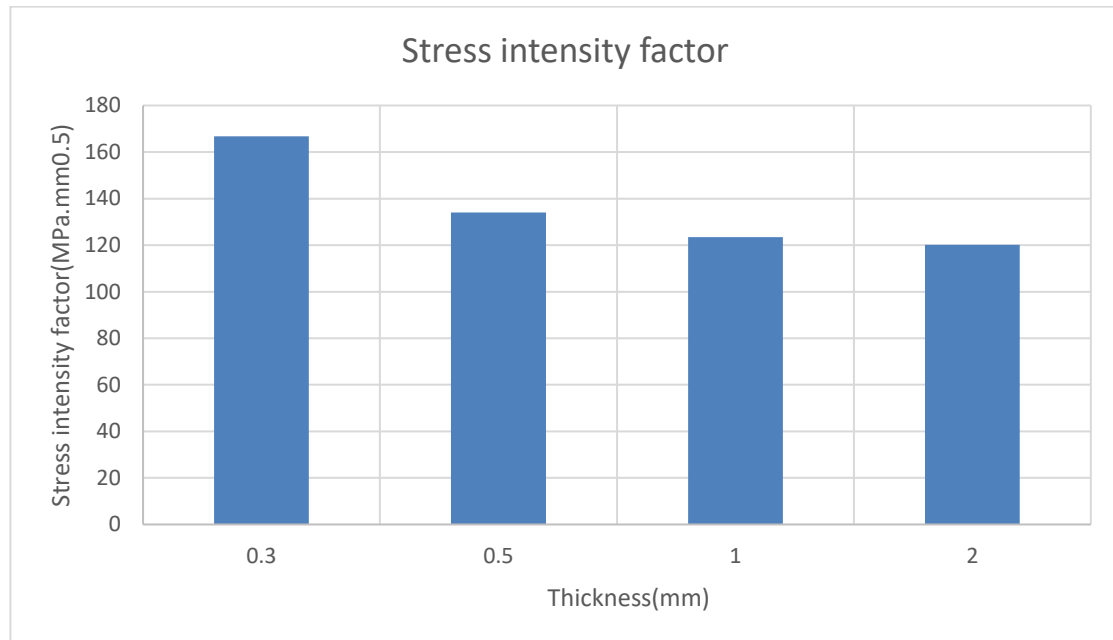


Figure 48: Thickness and Stress intensity factor graph

Stress intensity factors define the stress distribution near crack tips and are therefore important for the prediction of crack growth. The presented figure 48(Thickness and Stress intensity factor graph) shows correlation of plate thickness and SIFs,demonstrate that thinner plates possess appreciably higher SIFs, hence showing much more concentrated stress at the crack tip. Such behavior was also found in study of [39],where it was shown that thinner structures are less resistance to stress since they have reduced capabilities for load redistribution. Numerical approaches like XFEM permit the approximation of more accurate SIFs of these cases.

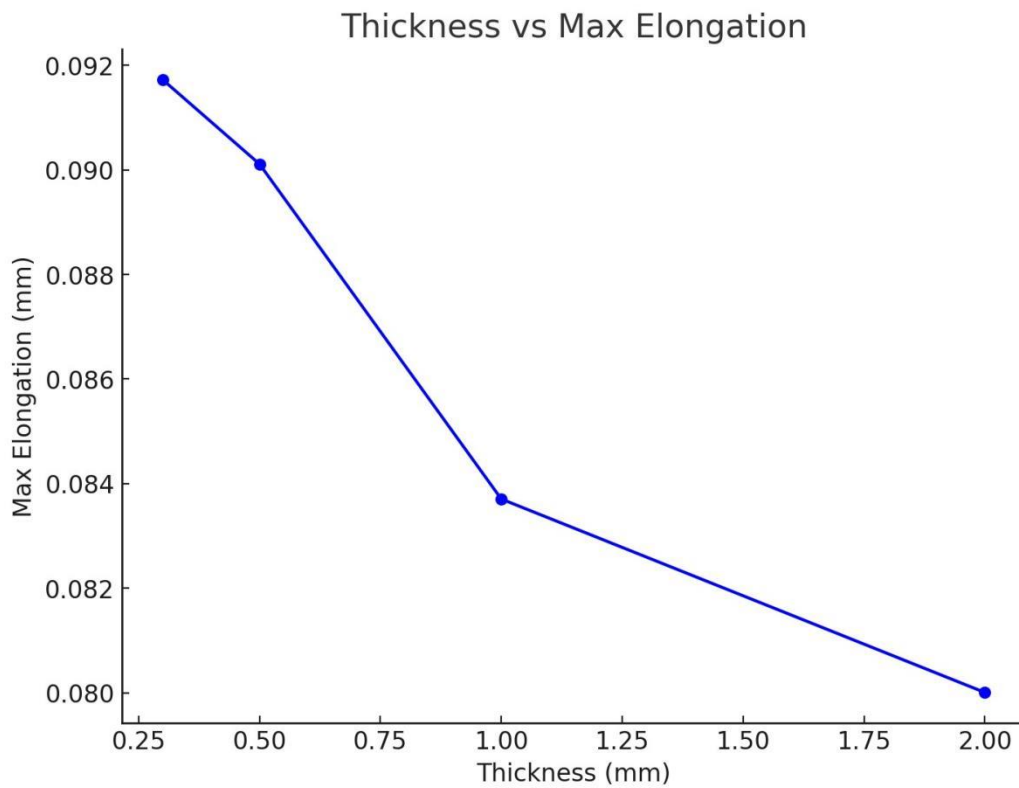


Figure 49: Thickness vs Max Elongation graph

Energy release rates are coupled with SIFs to quantify the energy availability toward crack prolongations ,thinner plates while having higher SIFs ,demonstrated increased elongation before failure as shown in figure 49.This indicates that thinner material can dissipate more energy up to a critical failure point, which agrees with the observation of [40].However , this higher elongation does not compensate for the crack growth observed in the material, again pointing out the balance between flexibility and fracture resistance.

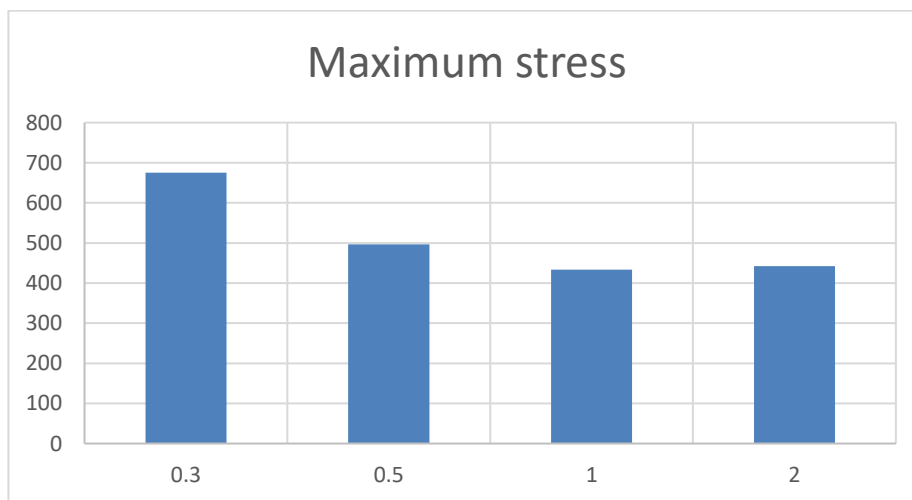


Figure 50:Thickness and Maximum stress graph

Figure 50, which outlines how thickness relates to maximum stress, shows the better toughness of a thicker plate. As mentioned by[41], higher thickness materials will have better stress distribution which slows the crack initiation and propagation process. This view coincides with the energy release rate since they are in similar conditions, the plate with higher thickness presents less driving force for crack development.

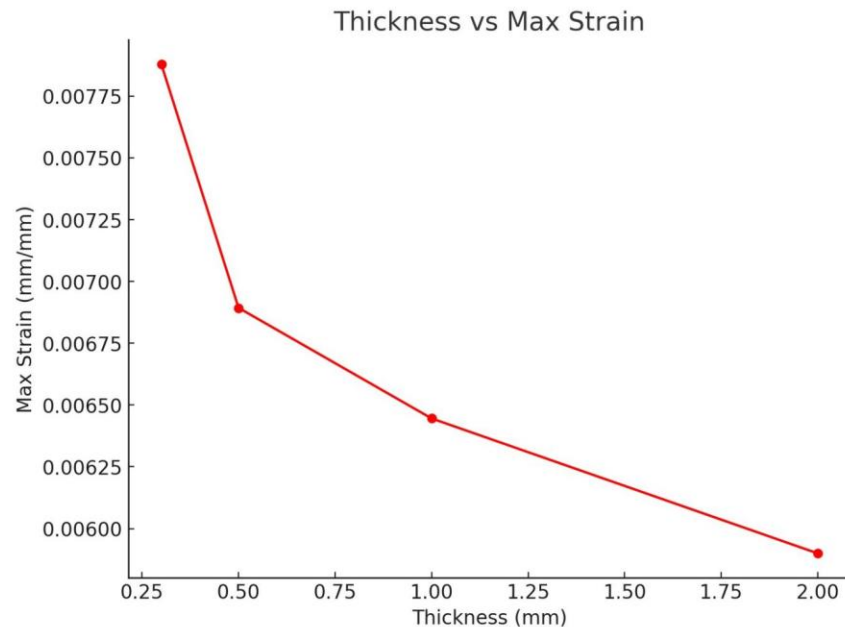


Figure 51: Thickness vs Max strain graph

Not only material thickness but also strain capacity affects crack propagation paths. The trend of thickness vs Maximum strain in the presented graph 51 is seen to be decreasing, showing that thicker plates have higher stiffness and will deform less under stress. This observation agrees with previous studies by[42], where it was observed that thicker composites have lower strain values and higher structural stability for different applied loads.

In summary, all these findings put forward the fact that SIFs, energy release rates, and crack propagation paths are interrelated in deciding the performance of materials. Though thinner plates allow more deformation, they undergo fast crack growth because of high stress concentration at the crack tip. While a thicker plate could resist higher stresses,

it becomes more capable of stress distribution. Therefore, it tends to show a better resistance to fracture. The insight arising from such a study underlines that an optimum balance between material property and thickness has to be looked for in an attempt to optimize the fracture behavior, keeping in view specific application requirements. Advances in computational techniques, including XFEM and cohesive zone modeling enable further enhancements of capability in the modeling of these cases in order to guide material design towards better structural reliability.

Conclusion

This report presents a comprehensive review and critical comparison between composite plates of different thicknesses and how important thickness is for fracture resistance under force. The thinner plates are not as good and have lower fracture toughness under force always and rather have increased concentration of stress around the crack tip which results in quicker and increased crack extension and tilt of the fracture. On the other side, thicker plates possess higher degree of fracture due to the fact that they can withstand greater forces acting on a larger volume hence the time taken for a crack to form or propagate will be longer due to the amount of energy consumed during the entire fracture process. This flexibility between thickness and function makes the necessity of bespoke designs of composites in aerospace, automotive and construction industries where weight is an important consideration.

The advanced simulation tools such as the Extended Finite Element Method (XFEM) and cohesive zone modelling were very helpful in understanding such processes. These techniques allow for exact simulation of both the processes of fracture creation as well as fracture extension making it possible to optimize composite plates for required performance parameters under use of the minimal amount of material. With consideration to the above findings, it is noted that there is a need to balance weight and fracture resistance of the designed composite materials. It is also recommended that future works in this area should focus on incorporation of new features such as phase-field models and artificial intelligence for better fracture prediction. Additionally, relying upon experimental data to confirm simulation results is important for reliability and thereby helping in the development of robust and application specific composite materials capable to satisfy present engineering problems.

References

- [1] Lu R, Li KS, Wang J, Yang J, Zhang XC, Tu ST. Understanding the small creep-fatigue crack growth mechanism of polycrystalline alloy based on crystal plasticity and extended finite element method. *Eng Fract Mech* 2024;306:110172. <https://doi.org/10.1016/J.ENGFRACMECH.2024.110172>.
- [2] Ravi Chandran KS, Galyon Dorman SE. The nature of specimen-size-effect on fatigue crack growth and net-section fracture mechanics approach to extract the size-independent behavior. *Int J Fatigue* 2021;145:106088. <https://doi.org/10.1016/J.IJFATIGUE.2020.106088>.
- [3] Cheng D, Gao Y, Li G. Fracture strength analysis of large-size and thin photovoltaic monocrystalline silicon wafers. *Eng Fract Mech* 2024;310:110523. <https://doi.org/10.1016/J.ENGFRACMECH.2024.110523>.
- [4] Thionnet A. From Fracture to Damage Mechanics: A behavior law for microcracked composites using the concept of Crack Opening Mode. *Compos Struct* 2010;92:780–94. <https://doi.org/10.1016/J.COMPSTRUCT.2009.08.047>.
- [5] Li J, Chang L, Zheng D, Wang Z, Zhang W, Ji Y, et al. In-situ experimental investigation of the small fatigue crack behavior in CP-Ti: The influence of loading parameters and directions. *Eng Fract Mech* 2025;314:110722. <https://doi.org/10.1016/J.ENGFRACMECH.2024.110722>.
- [6] Ren Z, Chang H, Yue Q, Liu X. FEA fatigue damage evolution model for stainless-steel-cruciform-joint utilizing AE-based damage mechanics. *J Constr Steel Res* 2025;226:109281. <https://doi.org/10.1016/J.JCSR.2024.109281>.
- [7] Zhang P, Liu C, Liu Y, Gu X. A friction energy-based damage model for discrete element simulation of fatigue damage evolution in concrete. *Constr Build Mater* 2024;456:139225. <https://doi.org/10.1016/J.CONBUILDMAT.2024.139225>.
- [8] Khan TA, Kim H, Kim HJ. Fatigue delamination of carbon fiber-reinforced polymer-matrix composites. *Failure Analysis in Biocomposites, Fibre-Reinforced Composites and Hybrid Composites* 2018:1–28. <https://doi.org/10.1016/B978-0-08-102293-1.00001-2>.
- [9] Usman M, Waheed S, Mubashar A. Effect of shape on void growth: A coupled Extended Finite Element Method (XFEM) and Discrete Dislocation Plasticity (DDP) study. *European Journal of Mechanics - A/Solids* 2022;92:104471. <https://doi.org/10.1016/J.EUROMECHSOL.2021.104471>.
- [10] Xu F, Hajibeygi H, Sluys LJ. Adaptive multiscale extended finite element method (MS-XFEM) for the simulation of multiple fractures propagation in geological formations. *J Comput Phys* 2023;486:112114. <https://doi.org/10.1016/J.JCP.2023.112114>.
- [11] Signorini C, Ahmed AH, Liebscher M, Zhao J, Köberle T, Mechtcherine V. Hybrid fibre-reinforced cementitious composites with short polyethylene and continue carbon fibres: Influence of roving impregnation on tensile and cracking behaviour. *Mater Des* 2024;248:113465. <https://doi.org/10.1016/J.MATDES.2024.113465>.
- [12] Němeček J, Čtvrtlík R, Václavek L, Němeček J. Fracture toughness of cement paste constituents assessed by micro-scratching correlated with acoustic emission. *Cem Concr Res* 2024;185:107623. <https://doi.org/10.1016/J.CEMCONRES.2024.107623>.
- [13] Wang P, Yang J, Liu W, Tang XZ, Zhao K, Lu X, et al. Tunable crack propagation behavior in carbon fiber reinforced plastic laminates with polydopamine and graphene oxide treated fibers. *Mater Des* 2017;113:68–75. <https://doi.org/10.1016/J.MATDES.2016.10.013>.
- [14] Han Z, Liu K, Zhou L, Fang F, Jiang J, Zhou X. Architecting unusual dual-gradient structures to overcome the strength-ductility trade-off in metallic materials. *J Mater Process Technol* 2025;335:118670. <https://doi.org/10.1016/J.JMATPROTEC.2024.118670>.
- [15] Gao D, Cai J, Liu Z, Wang Z, Guo X, Kang R. Optimization method research of ultrasonic assisted grinding processing for thin-wall reflectors of hard and brittle materials. *Ceram Int* 2024. <https://doi.org/10.1016/J.CERAMINT.2024.12.204>.
- [16] Hang ZY, Mi ZK, Wu YF, Wang JH, Yu Y. Fracture performance of concrete-epoxy mortar interface subjected to bending and shear: Experimental and numerical study. *Journal of Building Engineering* 2024;97:110887. <https://doi.org/10.1016/J.JOBE.2024.110887>.

- [17] Wang P, Yang J, Liu W, Tang XZ, Zhao K, Lu X, et al. Tunable crack propagation behavior in carbon fiber reinforced plastic laminates with polydopamine and graphene oxide treated fibers. *Mater Des* 2017;113:68–75. <https://doi.org/10.1016/J.MATDES.2016.10.013>.
- [18] Gonzalez Assias SL, Kotik HG, Perez Ipiña JE. Fracture surface analysis of fracture mechanics specimens with splits: Determination of the physical crack extension. *Theoretical and Applied Fracture Mechanics* 2024;133:104509. <https://doi.org/10.1016/J.TAFMEC.2024.104509>.
- [19] Srivastava VK, Gries T, Quadflieg T, Mohr B, Kolloch M, Kumar P. Fracture behavior of adhesively bonded carbon fabric composite plates with nano materials filled polymer matrix under DCB, ENF and SLS tests. *Eng Fract Mech* 2018;202:275–87. <https://doi.org/10.1016/J.ENGFRACMECH.2018.09.030>.
- [20] Fakoor M, Shahsavar S. Fracture assessment of cracked composite materials: Progress in models and criteria. *Theoretical and Applied Fracture Mechanics* 2020;105:102430. <https://doi.org/10.1016/J.TAFMEC.2019.102430>.
- [21] Ju M, Li X, Li X, Zhang G. A review of the effects of weak interfaces on crack propagation in rock: From phenomenon to mechanism. *Eng Fract Mech* 2022;263. <https://doi.org/10.1016/j.engfracmech.2022.108297>.
- [22] Azinpour E, Rzepa S, Melzer D, Reis A, Džugan J, Cesar de Sa JMA. Phase-field ductile fracture analysis of multi-materials and functionally graded composites through numerical and experimental methods. *Theoretical and Applied Fracture Mechanics* 2023;125:103906. <https://doi.org/10.1016/J.TAFMEC.2023.103906>.
- [23] Turis M, Ivánková O. Using finite element analysis to obtain plastic zones in the vicinity of the crack edges, under mixed mode loading conditions. *MATEC Web of Conferences* 2020;310:00028. <https://doi.org/10.1051/MATECCONF/202031000028>.
- [24] Pfeifer M. Degradation and Reliability of Materials. *Materials Enabled Designs* 2009:161–87. <https://doi.org/10.1016/B978-0-7506-8287-9.00006-9>.
- [25] Turis M, Ivánková O. Using finite element analysis to obtain plastic zones in the vicinity of the crack edges, under mixed mode loading conditions. *MATEC Web of Conferences* 2020;310:00028. <https://doi.org/10.1051/MATECCONF/202031000028>.
- [26] Alshoaibi AM, Fageehi YA. Advances in Finite Element Modeling of Fatigue Crack Propagation. *Applied Sciences* 2024;14:9297. <https://doi.org/10.3390/app14209297>.
- [27] Alshoaibi AM, Fageehi YA. Advances in Finite Element Modeling of Fatigue Crack Propagation. *Applied Sciences* 2024;14:9297. <https://doi.org/10.3390/app14209297>.
- [28] Ling D, Bu L, Tu F, Yang Q, Chen Y. A finite element method with mesh-separation-based approximation technique and its application in modeling crack propagation with adaptive mesh refinement. *Int J Numer Methods Eng* 2014;99:487–521. <https://doi.org/10.1002/nme.4689>.
- [29] Ling D, Bu L, Tu F, Yang Q, Chen Y. A finite element method with mesh-separation-based approximation technique and its application in modeling crack propagation with adaptive mesh refinement. *Int J Numer Methods Eng* 2014;99:487–521. <https://doi.org/10.1002/nme.4689>.
- [30] Kumar A, Shitole P, Ghosh R, Kumar R, Gupta A. Experimental and numerical comparisons between finite element method, element-free Galerkin method, and extended finite element method predicted stress intensity factor and energy release rate of cortical bone considering anisotropic bone modelling. *Proc Inst Mech Eng H* 2019;233:823–38. <https://doi.org/10.1177/0954411919853918>.
- [31] Kumar A, Shitole P, Ghosh R, Kumar R, Gupta A. Experimental and numerical comparisons between finite element method, element-free Galerkin method, and extended finite element method predicted stress intensity factor and energy release rate of cortical bone considering anisotropic bone modelling. *Proc Inst Mech Eng H* 2019;233:823–38. <https://doi.org/10.1177/0954411919853918>.
- [32] Li J, Leguillon D. Finite element implementation of the coupled criterion for numerical simulations of crack initiation and propagation in brittle materials. *Theoretical and Applied Fracture Mechanics* 2018;93:105–15. <https://doi.org/10.1016/j.tafmec.2017.07.010>.
- [33] Li J, Leguillon D. Finite element implementation of the coupled criterion for numerical simulations of crack initiation and propagation in brittle materials. *Theoretical and Applied Fracture Mechanics* 2018;93:105–15. <https://doi.org/10.1016/j.tafmec.2017.07.010>.

- [34] Mirdehghan SA. Fibrous polymeric composites. Engineered Polymeric Fibrous Materials, Elsevier; 2021, p. 1–58. <https://doi.org/10.1016/B978-0-12-824381-7.00012-3>.
- [35] Mirdehghan SA. Fibrous polymeric composites. Engineered Polymeric Fibrous Materials, Elsevier; 2021, p. 1–58. <https://doi.org/10.1016/b978-0-12-824381-7.00012-3>.
- [36] Radford DW, Teghtsoonian E. FRACTURE TOUGHNESS OF CARBON FIBRE/EPOXY COMPOSITES. TEQC83, Elsevier; 1983, p. 27–35. <https://doi.org/10.1016/B978-0-408-22162-7.50007-5>.
- [37] Radford DW, Teghtsoonian E. FRACTURE TOUGHNESS OF CARBON FIBRE/EPOXY COMPOSITES., 1983, p. 27–35. <https://doi.org/10.1016/b978-0-408-22162-7.50007-5>.
- [38] Campilho RDSG, Madani K, Belhouari M. Introduction to fracture mechanics. Fracture Mechanics: Advances in Research and Applications 2024;1–15. <https://doi.org/10.13140/RG.2.1.1444.2408>.
- [39] Tilbrook MT, Reimanis IE, Rozenburg K, Hoffman M. Effects of plastic yielding on crack propagation near ductile/brittle interfaces. Acta Mater 2005;53:3935–49. <https://doi.org/10.1016/j.actamat.2005.04.045>.
- [40] Wang B, Wang Y, Fan C, Zhang W. Energy distribution and evolution of frozen silty clay at subzero temperatures under compressive loading. Transportation Geotechnics 2021;31:100656. <https://doi.org/10.1016/j.trgeo.2021.100656>.
- [41] Sadowski AJ, Rotter JM, Reinke T, Ummenhofer T. Statistical analysis of the material properties of selected structural carbon steels. Structural Safety 2015;53:26–35. <https://doi.org/10.1016/j.strusafe.2014.12.002>.
- [42] Fan Q, Duan H, Xing X. A review of composite materials for enhancing support, flexibility and strength in exercise. Alexandria Engineering Journal 2024;94:90–103. <https://doi.org/10.1016/j.aej.2024.03.048>.